

## Low Output Current Ripple Weinberg Converter for Battery Discharge Regulator in Geostationary Satellites

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# Low Output Current Ripple Weinberg Converter for Battery Discharge Regulator in Geostationary Satellites

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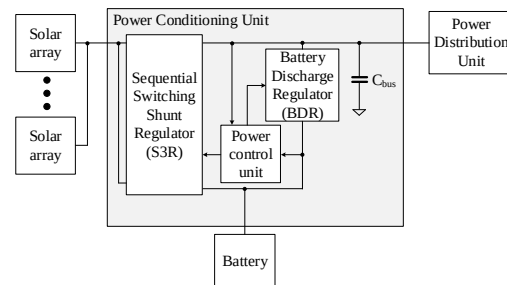
**Abstract**— The electrical power systems (EPSs) of geostationary orbit (GEO) satellites are typically designed with a regulated bus to provide stable power to the loads. In particular, when designing the EPS of a GEO satellite with a regulated bus, a battery discharge regulator (BDR) is necessary to supply the power required by the loads during eclipses (i.e., when the solar array does not generate any power). The conventional Weinberg converter is commonly employed in BDRs due to its continuous input and output currents, soft-switching operation of switches, lack of reverse recovery problems with diodes, and high efficiency. However, this converter has some drawbacks. For example, when all the switches are turned off, the output current rapidly increases, resulting in a large output current and voltage ripple, as well as a large rms current. This problem directly affects the lifetime and total capacitance of the bus capacitor. Moreover, this factor is directly related to the lifetime and size of the GEO satellite. Therefore, reducing the output current ripple of the BDR is an important design consideration. This paper proposes a converter that can reduce the output current ripple by adding the link capacitors and auxiliary windings of the coupled inductor. The proposed converter was validated with a prototype of 30-42 V input and 50 V output (750 W), operating at 100 kHz switching frequency. Additionally, the average efficiency of the proposed converter achieved approximately 97.5 %.

**Index Terms**—Battery discharge regulator, dc/dc converter, electrical power system, geostationary Earth orbit satellite, low output current ripple, Weinberg converter.

## I. INTRODUCTION

GEOSTATIONARY orbit (GEO) satellites are widely used for various purposes, such as communication, navigation, and weather observation. Recently, the increased demand for satellite communications has created a commensurate increasing demand for GEO satellites [1]–[4]. Additionally, as GEO satellites undertake more diverse and advanced missions, an improved electrical power system (EPS) with better quality and higher power is necessary for the various

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**Fig. 1** The sequential switching shunt regulator (S3R) & battery discharge regulator (BDR) topology for regulated bus voltage in the GEO satellite [22].

payload units [5]–[10]. However, any increase in the power of the EPS is closely related to an increase in its volume and weight [11], [12]. Therefore, it is inevitable that the cost of launching a GEO satellite will increase in accordance with the increased power requirements of the EPS [13]. Hence, it is essential to achieve high power density, light weight, and miniaturization in GEO satellites to accommodate the increased power requirement for payload units without incurring additional launch costs. Various recent research studies on these aspects have been conducted [14]–[21]. In particular, as referenced in [17], [18], reducing the volume and weight of the EPS is considered an effective approach. Usually, the EPS of a GEO satellite is designed with a regulated bus to supply stable power to the loads [22]–[24]. Fig. 1 displays the power conditioning unit (PCU) of the EPS for a GEO satellite, which consists of a sequential switching shunt regulator (S3R), a battery discharge regulator (BDR), a power control unit, and a bus capacitor for the regulated bus. As demonstrated in Fig. 1, the PCU can precisely and stably control the bus voltage through the BDR, even when power is not generated from the solar array. Specifically, the bus capacitor in the PCU is generally designed to be very large to maintain a stable bus voltage. Therefore, reducing the required capacitance of the bus capacitor is also important for implementing a high power density and lightweight EPS.

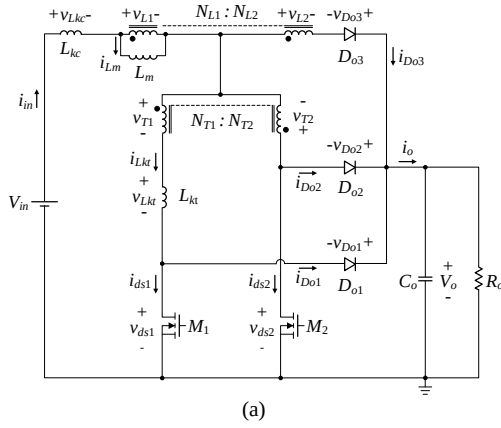
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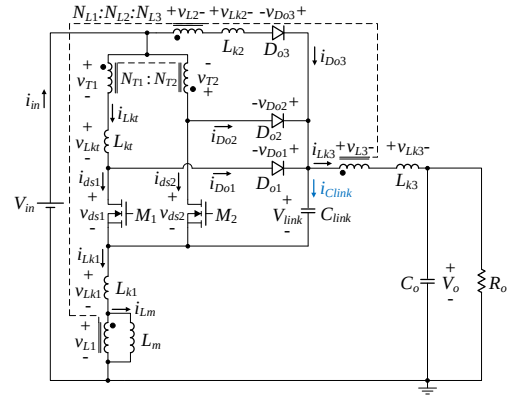
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**Fig. 2** Circuit diagram and key operation waveforms of the conventional Weinberg converter (a) Circuit diagram of the conventional Weinberg converter (b) Key waveforms of the current.

In general, the BDR shown in Fig. 1 requires a step-up converter that can achieve continuous input and output current, low noise, and high efficiency. One way to achieve high efficiency is the soft switching operation of the switch. When operating with soft switching, current and voltage do not overlap significantly when the switch is turned on, which can reduce switching losses and is advantageous for achieving high efficiency. Research on soft switching operation is also continuously ongoing [25], [26]. Many different topologies have been proposed for BDR applications [27]–[38]. Boost converters with a lossless snubber can reduce the reverse recovery loss of the diode and achieve high-frequency operation due to the small saturable inductor [28]. However, the converter proposed in [28] has the disadvantage of placing a burden on the output capacitor due to the discontinuous output current. In contrast, the converter in [29] achieved continuous output current and small output current ripple; although it suffered from poor noise characteristics due to the hard-switching operation of switches and the reverse recovery problem of the diode. The circuits suggested in [30]–[33] had continuous input and output currents, enabling soft-switching operation. However, they had the disadvantage of added complexity due to the additional circuit required to achieve soft-switching operation. As an alternative to the mentioned topologies, the conventional Weinberg converter shown in



**Fig. 3** The circuit diagram of the proposed converter.

Fig. 2(a) has been widely used recently [34]–[38] and is well known for the following advantages that render it a suitable topology for BDRs:

- The input and output currents are continuous.
- The operating frequency of the converter is twice the switching frequency, allowing for a reduction in the size of the magnetic components.
- Soft-switching operation of the switches in the converter is possible across the entire input voltage and output load ranges.
- Since the voltage across the switches is clamped to the output voltage, the snubber is not required.

However, as shown in Fig. 2(b), the conventional Weinberg converter has some drawbacks. When all the switches are turned off, the output current increases sharply, causing a significant increase in the output current ripple. In turn, this causes an increase in both the rms current flowing through the output capacitor and the output voltage ripple. Moreover, these problems become increasingly critical as the load current increases. Thus, to solve these problems, a significant number of output capacitors are required. However, the use of a large number of output capacitors increases the size and weight of the EPS.

To address the previously mentioned disadvantages of conventional converters, a converter with a link capacitor and an additional tertiary winding of the coupled inductor is proposed in this paper, as shown in Fig. 3. More specifically, the proposed converter can reduce the output current ripple with the help of the coupled inductor when all switches are turned off. Additionally, as the load current increases, the proposed converter exhibits a significantly smaller increase in output current ripple compared to conventional converters. Therefore, the number of bus capacitors in the PCU (which is the output capacitor of the BDR) can be significantly reduced. Furthermore, leakage inductors in the proposed converter can prevent reverse recovery problems of all diodes and reduce turn-on switching losses of all the switches by guaranteeing soft-switching operation. This results in higher efficiency and lower output side noise. As a result, the noise on the converter's output side decreased for all of the satellite's payload units.

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TABLE I  
COMPARISON OF TOPOLOGIES

|                             | [36]  | [39]  | [40]  | [41]  | Proposed converter |
|-----------------------------|-------|-------|-------|-------|--------------------|
| Number of the diode         | 3     | 1     | 1     | 3     | 3                  |
| Number of the switch        | 2     | 1     | 1     | 2     | 2                  |
| Number of the magnetic core | 2     | 2     | 2     | 3     | 2                  |
| Number of the capacitor     | 2     | 2     | 2     | 2     | 2                  |
| Soft-switching operation    | O     | X     | X     | O     | O                  |
| Diode reverse recovery      | X     | O     | O     | X     | X                  |
| Peak current of the switch  | 14.1A | 30.9A | 56.8A | 15A   | 15.4A              |
| Peak current of the diode   | 14.1A | 30.9A | 56.8A | 15A   | 15.4A              |
| Output current ripple       | 15.5A | 10.2A | 21.2A | 4.26A | 6.23A              |
| Output voltage ripple       | 0.41V | 0.83V | 1.72V | 0.17V | 0.29V              |

Table I shows the comparison results between several converters with continuous input and output currents and the proposed converter, under the same specifications ( $V_{in} = 36$  V,  $V_o = 50$  V,  $P_o = 750$  W,  $f_{sw} = 100$  kHz). The inductance is set to 10  $\mu$ H, and the output capacitance is set to 15.4  $\mu$ F for the comparison. As shown in Table I, [39] and [40] have the disadvantage of operating with hard switching and facing the issue of diode reverse recovery, despite the advantage of using fewer semiconductor components. Moreover, the large peak current flowing through each device poses limitations for use in high-current applications. In contrast, [36], [41], and proposed converter, although using a larger number of semiconductor devices compared to the two previously mentioned converters, have the advantage of being capable of soft switching operation and avoiding diode reverse recovery issues. Additionally, the peak current flowing through the semiconductor devices is relatively small, making them advantageous for high-current applications. However, [36] has a drawback that the output voltage ripple is larger compared to [41] and the proposed converter due to the larger output current ripple. The proposed

converter, as well as the one mentioned in reference [41], exhibits significantly lower output voltage and current ripples compared to other converters. Although the proposed converter and [41] share a similar topology to reduce output current ripple, the proposed converter achieves this with one less magnetic device compared to [41]. Additionally, the methods employed for optimizing the converter design are also different. Furthermore, since BDRs are generally used in parallel with multiple converters, these differences in weight and volume become even greater. Therefore, it can be observed from Table I that the proposed converter among the topologies presented is the most suitable for satellite applications.

On the other hand, one effective method to significantly reduce the output voltage ripple of the conventional Weinberg converter is to add a post LC filter to the output side. However, to add an LC filter, additional magnetic devices capable of handling high load currents need to be included. Moreover, due to the continuous increase in load current with the ongoing rise in satellite power, the burden on the magnetic components constituting the LC filter will also increase. In contrast, the proposed converter achieves the benefits of adding an LC filter without the need for additional magnetic devices by adding auxiliary windings to the coupled inductor. Therefore, the proposed converter has a significant advantage in reducing the output current ripple without requiring additional magnetic devices to handle large load currents.

## II. OPERATIONAL PRINCIPLES

The proposed converter operates with two switches ( $M_1$  and  $M_2$ ) alternating within the same duty cycle. To facilitate analysis of the converter's operation, the following assumptions are made:

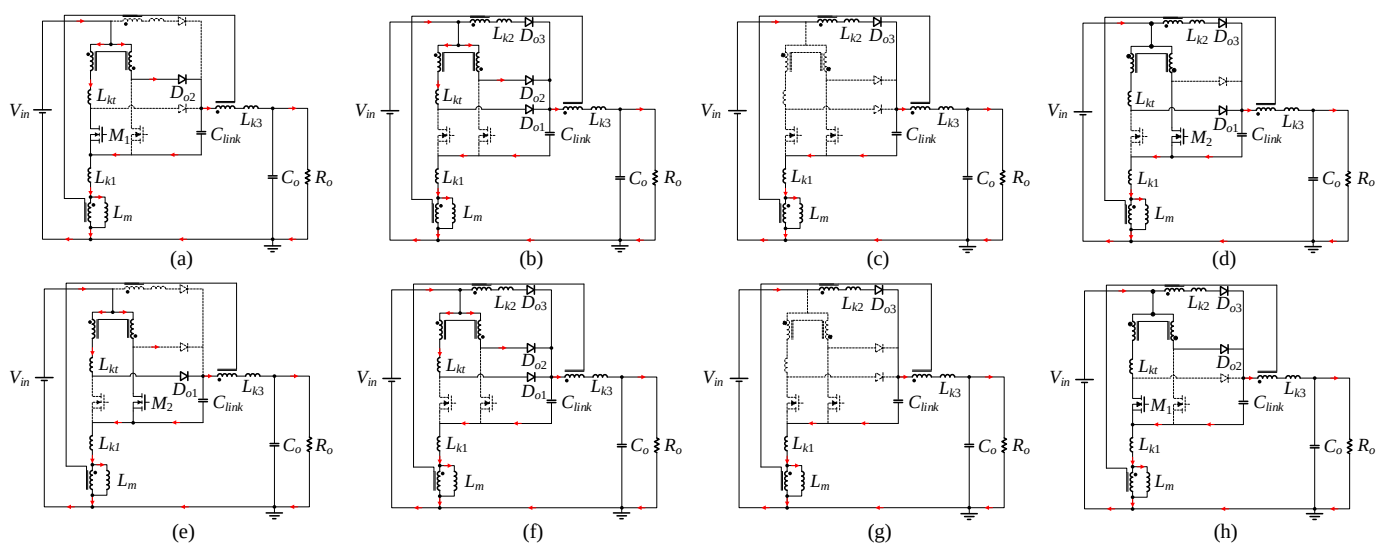


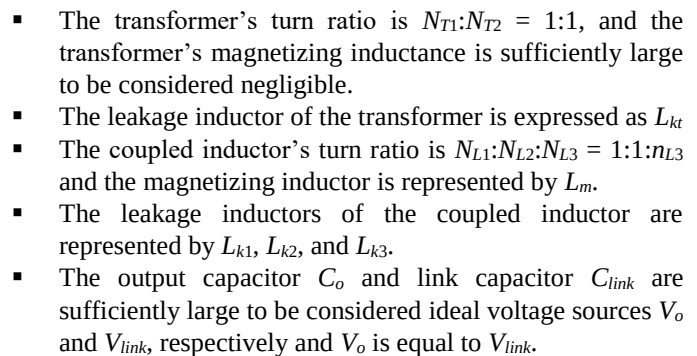
Fig. 4 The circuit operation of the proposed converter during one period (a) Mode 1 ( $t_1$ - $t_2$ ), (b) Mode 2 ( $t_2$ - $t_3$ ), (c) Mode 3 ( $t_3$ - $t_4$ ), (d) Mode 4 ( $t_4$ - $t_5$ ), (e) Mode 5 ( $t_5$ - $t_6$ ), (f) Mode 6 ( $t_6$ - $t_7$ ), (g) Mode 7 ( $t_7$ - $t_8$ ), (h) Mode 8 ( $t_8$ - $t_9$ ).

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TABLE II

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**Fig. 6** Key operation waveforms of proposed converter.

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**Mode 1 [ $t_1$ - $t_2$ ]:** This mode starts after the commutation between diodes  $D_{o2}$  and  $D_{o3}$  is completed and  $D_{o3}$  is blocked. From Fig. 5(a) and Table II, the voltage across the coupled inductor  $v_{L1}$  ( $=v_{L2}=v_{L3}/n_{L3}$ ) can be derived as follows:

$$v_{L1} = \frac{L_m}{L_{md1} + L_m} v_{md1}. \quad (1)$$

According to Kirchhoff's current law (KCL),  $i_{Lk1} + i_{Lk3} = 2i_{Lkt}$  and  $i_{Lk1} + n_{L3}i_{Lk3} = i_{Lm}$ . In addition,  $v_{Lk1} = V_{in} + v_{T2} - V_o - v_{L1}$ ;  $v_{Lk3} = V_{in} + v_{T2} - v_{L3} - V_o$  and  $v_{Lkt} = V_o - v_{T1} - v_{T2}$  by Kirchhoff's voltage law (KVL). Therefore,  $v_{T1}$  ( $=v_{T2}$ ) can be obtained as follows:

$$v_{T1} = L_{pal} v_{L1} + V_o - V_{in} \quad (2)$$

where  $L_{pal} = (L_m L_{k3} + n_{L3}^2 L_{k1} L_m + L_{k1} L_{k3}) / (n_{L3} L_m L_{k1} + L_m L_{k3})$ . Using the equations derived from KVL and KCL previously in addition to (1) and (2), the voltages across  $L_{k1}$ ,  $L_{k3}$ , and  $L_{kt}$  are as follows:

$$v_{Lk1} = \frac{n_{L3}(n_{L3}-1)L_m L_{k1} + L_{k1} L_{k3}}{L_m(n_{L3} L_{k1} + L_{k3})} v_{L1} \quad (3)$$

$$v_{Lk3} = \frac{(1-n_{L3})L_m L_{k1} + L_{k1} L_{k3}}{n_{L3} L_m L_{k1} + L_m L_{k3}} v_{L1} \quad (4)$$

$$v_{Lkt} = 2V_{in} - V_o - 2L_{pal} v_{L1}. \quad (5)$$

In this mode, since  $M_1$  and  $D_{o2}$  are turned on,  $v_{Do1}$  and  $v_{ds2}$  are clamped to  $V_o$ . Therefore, from KVL,  $v_{Do3} = v_{L2} + v_{T2}$ . Hence,  $v_{Do3}$  can be expressed as follows using (1) and (2):

$$v_{Do3} = (L_{pal} + 1) v_{L1} + V_o - V_{in}. \quad (6)$$

From (1)–(5), the current flowing through  $L_m$ ,  $L_{k1}$ ,  $L_{k3}$ , and  $L_{kt}$  can be obtained as follows:

$$i_{Lm}(t) = \frac{v_{md1}}{L_{md1} + L_m} (t - t_1) + i_{Lm}(t_1) \quad (7)$$

$$i_{Lk(i)}(t) = \frac{v_{Lk(i)}}{L_{k(i)}} (t - t_1) + i_{Lk(i)}(t_1). \quad (8)$$

From (8), the currents flowing through  $L_{k1}$ ,  $L_{k3}$ , and  $L_{kt}$  correspond to  $i = 1, 3$ , and  $t$ , respectively.

**Mode 2 [ $t_2$ - $t_3$ ]:** This mode begins when  $M_1$  is turned off at  $t_2$ . During this mode,  $v_{ds1}$  increases from zero to reach  $V_{link} = V_o$ . When  $v_{ds1}$  reaches  $V_{link} = V_o$ ,  $v_{ds1}$  is clamped to  $V_{link} = V_o$  and  $D_{o1}$  and  $D_{o3}$  are turned on. From Fig. 5(b) and Table II, the voltage  $v_{L1}$  ( $=v_{L2}=v_{L3}/n_{L3}$ ) across the coupled inductor can be derived as follows:

$$v_{L1} = \frac{L_m}{L_{md2} + L_m} v_{md2}. \quad (9)$$

Since all diodes are turned on in this mode, from KCL and KVL,  $i_{Lk1} + i_{Lk3} = 2i_{Lkt} + i_{Lk2}$ ,  $i_{Lk1} + i_{Lk2} + n_{L3}i_{Lk3} = i_{Lm}$ ,  $v_{Lk1} = V_{in} + v_{T2} - V_o - v_{L1}$ ,  $v_{Lk2} = -v_{L2} - v_{T2}$ ,  $v_{Lk3} = V_{in} + v_{T2} - v_{L3} - V_o$ , and  $v_{Lkt} = -v_{T1} - v_{T2}$ . Therefore,  $v_{T1}$  ( $=v_{T2}$ ) can be derived by (9) and Table II:

$$v_{T1} = L_{cb2} \left( 1 + \frac{L_{k3}}{L_{k1}} \right) (V_o - V_{in}) + n_{L3} L_{cb2} v_{L1}. \quad (10)$$

The voltages across  $L_{k1}$ ,  $L_{k2}$ ,  $L_{k3}$ , and  $L_{kt}$  can be expressed as follows using (9), (10), and Table II:

$$v_{Lk1} = \left( \frac{L_{cb2} L_{k3}}{L_{k1}} + L_{cb2} - 1 \right) (V_o - V_{in}) + (n_{L3} L_{cb2} - 1) v_{L1} \quad (11)$$

$$v_{Lk2} = L_{cb2} \left( 1 + \frac{L_{k3}}{L_{k1}} \right) (V_o - V_{in}) - (n_{L3} L_{cb2} + 1) v_{L1} \quad (12)$$

$$v_{Lk3} = \left( \frac{L_{cb2} L_{k3}}{L_{k1}} + L_{cb2} - 1 \right) (V_o - V_{in}) + n_{L3} (L_{cb2} - 1) v_{L1} \quad (13)$$

$$v_{Lkt} = 2L_{cb2} \left( 1 + \frac{L_{k3}}{L_{k1}} \right) (V_{in} - V_o) - 2n_{L3} L_{cb2} v_{L1}. \quad (14)$$

The current flowing through  $L_m$ ,  $L_{k1}$ ,  $L_{k2}$ ,  $L_{k3}$ , and  $L_{kt}$  can be derived as follows by (9)–(14):

$$i_{Lm}(t) = \frac{v_{md2}}{L_{md2} + L_m} (t - t_2) + i_{Lm}(t_2) \quad (15)$$

$$i_{Lk(i)}(t) = \frac{v_{Lk(i)}}{L_{k(i)}} (t - t_2) + i_{Lk(i)}(t_2). \quad (16)$$

From (16), the currents flowing through  $L_{k1}$ ,  $L_{k2}$ ,  $L_{k3}$ , and  $L_{kt}$  correspond to  $i = 1, 2, 3$ , and  $t$ , respectively.

As shown in Fig. 6, commutation between  $D_{o1}$ ,  $D_{o2}$ , and  $D_{o3}$  occurs during  $t_2$ – $t_3$ , in this mode due to leakage inductances in the circuit. In addition, as shown in Fig. 4(b),  $i_{Do1}$  and  $i_{Do2}$  are the same as  $i_{Lkt}$  and  $i_{Do3}$  is the same as  $i_{Lk2}$ . During the commutation operation,  $i_{Do1}$  and  $i_{Do2}$  gradually decrease and  $i_{Do3}$  gradually increases according to (9), (12), (14), (16), and Table II. This commutation operation ends when  $i_{Do1}$  and  $i_{Do2}$  gradually decrease and reach zero. Then,  $D_{o1}$  and  $D_{o2}$  are softly turned off. Therefore, there are no reverse recovery problems for  $D_{o1}$  and  $D_{o2}$ . Specifically, unlike conventional converters, the output current ripple does not increase significantly in the proposed converter due to its structure and  $L_{k3}$  of the auxiliary winding.

**Mode 3 [ $t_3$ - $t_4$ ]:** Since the diode commutation operation has ended,  $D_{o1}$  and  $D_{o2}$  are blocked and  $D_{o3}$  is still conducting. From Fig. 5(c) and Table II, the voltage  $v_{L1}$  ( $=v_{L2}=v_{L3}/n_{L3}$ ) across the coupled inductor can be derived as follows:

$$v_{L1} = \frac{L_m}{L_{md3} + L_m} v_{md3}. \quad (17)$$

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Furthermore, from KCL and KVL,  $i_{Lk1} + i_{Lk3} = i_{Lk2}$ ,  $i_{Lm} = i_{Lk1} + i_{Lk2} + n_{L3}i_{Lk3}$ ,  $v_{Lk1} + v_{Lk2} = V_{in} - V_o - v_{L1} - v_{L2}$ , and  $v_{Lk2} + v_{Lk3} = V_{in} - V_o - (n_{L3} + 1)v_{L1}$ . Therefore, the voltages across  $L_{k1}$ ,  $L_{k2}$ , and  $L_{k3}$  can be expressed as follows:

$$v_{Lk1} = \frac{L_{k3}}{n_{L3}L_{k1}}(V_o - V_{in}) + \left( \frac{L_{k3}}{n_{L3}L_m} + \frac{2L_{k3}}{n_{L3}L_{k1}} + n_{L3} - 1 \right) v_{L1} \quad (18)$$

$$v_{Lk2} = \left( 1 + \frac{L_{k3}}{n_{L3}L_{k1}} \right) (V_{in} - V_o) - \left( \frac{L_{k3}}{n_{L3}L_m} + \frac{2L_{k3}}{n_{L3}L_{k1}} + n_{L3} + 1 \right) v_{L1} \quad (19)$$

$$v_{Lk3} = \frac{L_{k3}}{n_{L3}L_{k1}}(V_o - V_{in}) + \left( \frac{L_{k3}}{n_{L3}L_m} + \frac{2L_{k3}}{n_{L3}L_{k1}} \right) v_{L1}. \quad (20)$$

From (17)–(20), the current flowing through  $L_m$ ,  $L_{k1}$ ,  $L_{k2}$ , and  $L_{k3}$  can be obtained as follows:

$$i_{Lm}(t) = \frac{v_{md3}}{L_{md3} + L_m}(t - t_3) + i_{Lm}(t_3) \quad (21)$$

$$i_{Lk(i)}(t) = \frac{v_{Lk(i)}}{L_{k(i)}}(t - t_3) + i_{Lk(i)}(t_3). \quad (22)$$

From (22), The currents flowing through  $L_{k1}$ ,  $L_{k2}$ , and  $L_{k3}$  are  $i = 1, 2$ , and  $3$ , respectively.

**Mode 4 [ $t_4$ – $t_5$ ]:** This mode starts when  $M_2$  is turned on. Since  $M_2$  is turned on,  $v_{D_{o2}}$  is clamped to  $V_{link} = V_o$ , and  $D_{o1}$  is turned on. From Fig. 5(d) and Table II, the voltage across the coupled inductor  $v_{L1}$  ( $=v_{L2}=v_{L3}/n_{L3}$ ) can be derived as follows:

$$v_{L1} = \frac{L_m}{L_{md4} + L_m} v_{md4}. \quad (23)$$

Furthermore, from KCL and KVL,  $i_{Lk1} + i_{Lk3} = 2i_{Lkt} + i_{Lk2}$ ,  $i_{Lm} = i_{Lk1} + i_{Lk2} + n_{L3}i_{Lk3}$ ,  $v_{Lk1} = V_{in} + v_{T2} - v_{L1}$ ,  $v_{Lk2} = -v_{L2} - v_{T2} - V_o$ ,  $v_{Lk3} = V_{in} + v_{T2} - v_{L3}$ , and  $v_{Lkt} = -v_{T1} - v_{T2} - V_o$ . Therefore,  $v_{T1}$  ( $=v_{T2}$ ) can be derived as follows from (23) and Table II:

$$v_{T1} = L_{k3}L_{cb4} \left( \frac{n_{L3}v_{L1} - V_{in}}{L_{k3}} - \frac{V_{in} + V_o}{L_{k1}} - \frac{2V_o}{L_{kt}} \right). \quad (24)$$

Through (23) and (24) and referencing Table II, the voltages across  $L_{k1}$ ,  $L_{k2}$ ,  $L_{k3}$ , and  $L_{kt}$  can be represented as follows:

$$v_{Lk1} = (n_{L3}L_{cb4} - 1)v_{L1} + \left( 1 - \frac{L_{k3} \parallel L_{cb4}}{L_{k1} \parallel L_{k3}} \right) V_{in} + \frac{L_{k3}L_{cb4}}{L_{k1} \parallel L_{kt}} V_o \quad (25)$$

$$v_{Lk2} = \frac{L_{k3} \parallel L_{cb4}}{L_{k1} \parallel L_{k3}} V_{in} + \left( \frac{L_{k3}L_{cb4}}{L_{k1} \parallel L_{kt}} - 1 \right) V_o - (1 + n_{L3}L_{cb4})v_{L1} \quad (26)$$

$$v_{Lk3} = (L_{cb4} - 1)n_{L3}v_{L1} + \left( 1 - \frac{L_{k3} \parallel L_{cb4}}{L_{k1} \parallel L_{k3}} \right) V_{in} - \frac{L_{k3}L_{cb4}}{L_{k1} \parallel L_{kt}} V_o \quad (27)$$

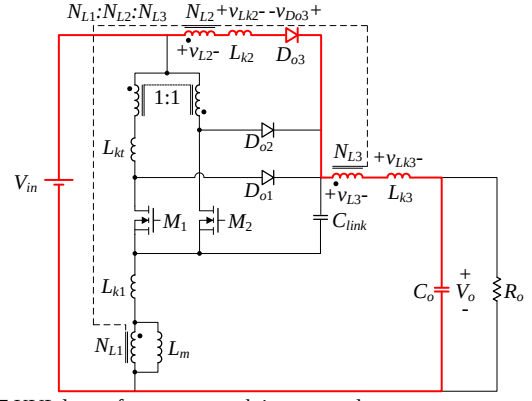


Fig. 7 KVL loop of outermost path in proposed converter.

$$v_{Lkt} = 2L_{k3}L_{cb4} \left( \frac{V_{in} + V_o}{L_{k1}} + \frac{2V_o}{L_{kt}} + \frac{V_{in} - n_{L3}v_{L1}}{L_{k3}} \right) - V_o. \quad (28)$$

Based on (23)–(28), the currents flowing through  $L_m$ ,  $L_{k1}$ ,  $L_{k2}$ ,  $L_{k3}$ , and  $L_{kt}$  can be obtained as follows:

$$i_{Lm}(t) = \frac{v_{md4}}{L_{md4} + L_m}(t - t_4) + i_{Lm}(t_4) \quad (29)$$

$$i_{Lk(i)}(t) = \frac{v_{Lk(i)}}{L_{k(i)}}(t - t_4) + i_{Lk(i)}(t_4). \quad (30)$$

From (30), the current flowing through  $L_{k1}$ ,  $L_{k2}$ ,  $L_{k3}$ , and  $L_{kt}$  corresponds to  $i=1, 2, 3$ , and  $t$ , respectively.

In this mode, commutation between  $D_{o1}$  and  $D_{o3}$  occurs during  $t_4$ – $t_5$ , as shown in Fig. 6. In addition, as shown in Fig. 4(d),  $i_{D_{o1}}$  and  $i_{ds2}$  are equal to  $i_{Lkt}$  and  $i_{D_{o3}}$  is equal to  $i_{Lk2}$ . During the commutation operation,  $i_{D_{o1}}$  and  $i_{ds2}$  gradually increase and  $i_{D_{o3}}$  gradually decreases according to (23), (26), (28), (30), and Table II. Therefore, soft-switching operation for  $M_2$  is achieved in this mode. Moreover, since  $i_{D_{o3}}$  gradually decreases and becomes zero during the commutation operation,  $D_{o3}$  is softly turned off. Therefore, there is no issue with reverse recovery for  $D_{o3}$ . Additionally, since the operation of this mode is similar to the operation during  $t_8$ – $t_9$  shown in Fig. 6, soft-switching operation is also possible when  $M_1$  is turned on. Consequently, turn-on losses of the switches can be significantly reduced. This mode ends when the commutation between  $D_{o1}$  and  $D_{o3}$  is completed. Because the operations during  $t_5$ – $t_9$  shown in Fig. 6 are similar to those of modes 1 to 4, a detailed analysis of their operation is omitted.

### III. ANALYSIS OF THE PROPOSED CONVERTER

#### A. Voltage conversion ratio

To determine the voltage conversion ratio of the proposed converter, KVL was applied to the outermost loop of the proposed converter, as shown in Fig. 7. Since the average voltage of the inductor is 0 V in steady-state, the average voltage  $\langle v_{D_{o3}} \rangle$  of  $D_{o3}$  is equal to  $V_o - V_{in}$ . In addition, since  $D_{o3}$  conducts for  $(1-2D)T_s + 2T_{CM(on)}$  within one switching cycle (as demonstrated in Fig. 6),  $\langle v_{D_{o3}} \rangle$  is derived as follows by (6):

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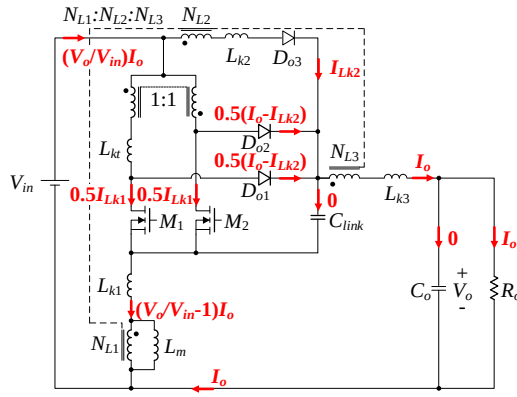


Fig. 8 Average currents through proposed converter circuit components

$$\langle v_{Do3} \rangle = 2D_{eff} (L_{pal} + 1) \frac{L_m v_{md1}}{L_{md1} + L_m} + 2D_{eff} (V_o - V_{in}). \quad (31)$$

From (31) and Table II, the effective duty cycle  $D_{eff}$  and the voltage conversion ratio can be obtained as follows:

$$D_{eff} = \frac{(L_{md1} + L_m)(V_o - V_{in})}{2L_m(1 + L_{pal})v_{md1} + 2(L_{md1} + L_m)(V_o - V_{in})} \quad (32)$$

$$\frac{V_o}{V_{in}} = \frac{1 + 2D_{eff}(2L_{eq} - 1)}{1 + 2D_{eff}(L_{eq} - 1)} \quad (33)$$

where  $L_{eq}$  is defined as follows:

$$L_{eq} = \frac{4L_m L_{k3} + 2L_{k1} L_{k3} + 2L_{k1} L_m (n_{L3}^2 + n_{L3})}{(L_m + L_{md1}) \{ (n_{L3} - 1) 2L_{kt} + 4(n_{L3}^2 L_{k1} + L_{k3}) \}}. \quad (34)$$

Since the conventional and proposed converters operate similarly, their voltage conversion ratios and effective duty cycles are similar.

### B. Commutation time of the diode

As shown in Fig. 6, when  $M_1$  or  $M_2$  is turned on, diode commutation is performed during  $T_{CM(on)}$ . In contrast, when  $M_1$  or  $M_2$  is turned off, diode commutation is performed during  $T_{CM(off)}$ . In particular,  $T_{CM(on)}$  can be derived from the current flowing through  $M_1$ , and Fig. 8 shows the average current flowing through each component of the circuit. For convenient analysis, the converter is assumed to be ideal except for the leakage inductors. As portrayed in Fig. 8, the average current flowing through  $M_1$  is  $0.5I_o(V_o/V_{in} - 1)$  by KCL. Therefore, the average current  $I_{ds1}$  of  $M_1$  can be derived based on Fig. 6 and mode analysis, as follows:

$$I_{ds1} = \left( \frac{v_{Lkt\_md4}}{2L_{kt}} \frac{T_{CM(on)}}{T_s} + \frac{v_{Lkt\_md4}}{L_{kt}} D_{eff} \right) T_{CM(on)} + \frac{v_{Lkt\_md1}}{2L_{kt}} D_{eff}^2 T_s. \quad (35)$$

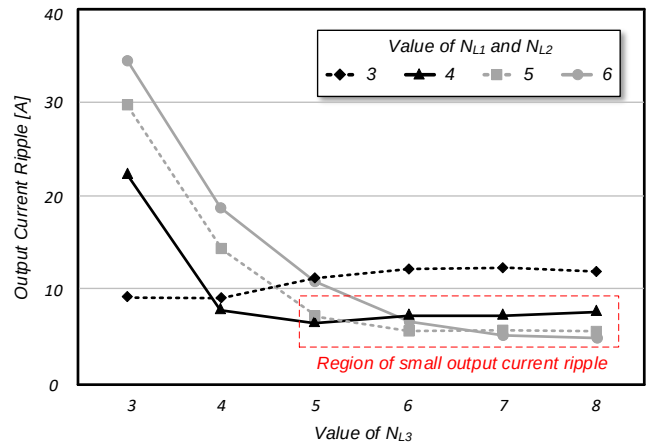


Fig. 9 Output current ripple according to  $N_{L1} (=N_{L2})$  and  $N_{L3}$  ( $V_{in} = 36$  V,  $V_o = 50$  V,  $P_o = 750$  W,  $L_{kt} = 0.3$  uH)

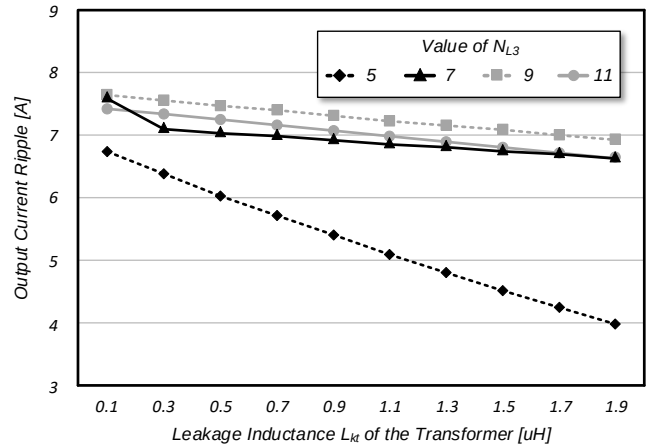


Fig. 10 The output current ripple according to  $N_{L3}$  and  $L_{kt}$  ( $V_{in} = 36$  V,  $V_o = 50$  V,  $P_o = 750$  W,  $N_{L1} = N_{L2} = 4$ )

Here,  $v_{Lkt\_md1}$  and  $v_{Lkt\_md4}$  represent the voltage across  $L_{kt}$  during Modes 1 and 4, respectively. Therefore,  $T_{CM(on)}$  can be expressed as follows using (35) and Fig. 8:

$$T_{CM(on)} = -D_{eff} T_s + \sqrt{D_{eff}^2 T_s^2 \left( 1 - \frac{v_{Lkt\_md1}}{v_{Lkt\_md4}} \right) + \frac{L_{kt} T_s I_o}{v_{Lkt\_md4}} \left( \frac{V_o}{V_{in}} - 1 \right)} \quad (36)$$

As demonstrated in Fig. 6,  $i_{Lkt}$  ( $=i_{Do1}=i_{Do2}$ ) decreases and becomes zero during the period from  $t_2$  to  $t_3$ . Therefore,  $T_{CM(off)}$  can be expressed as follows based on Fig. 6:

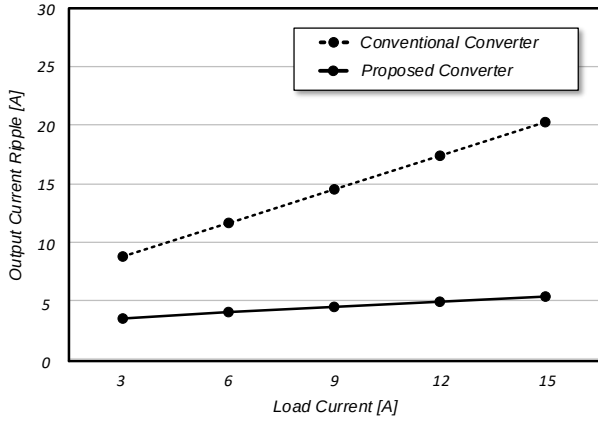
$$T_{CM(off)} = \frac{v_{Lkt\_md4} T_{CM(on)} + v_{Lkt\_md1} D_{eff} T_s}{v_{Lkt\_md2}} \quad (37)$$

Where  $v_{Lkt\_md2}$  represents the voltage across  $L_{kt}$  during Mode 2.

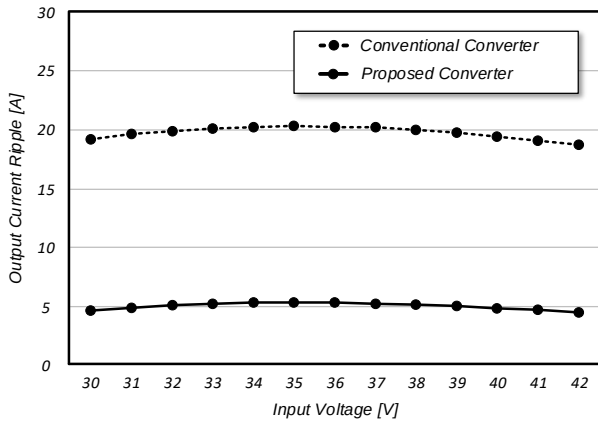
### C. Output current ripple

As shown in Fig. 3, the output current of the proposed converter is equal to  $i_{Lk3}$ . Furthermore, the voltage applied to  $L_{k3}$  for each mode was derived in Section II, and the effective duty

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**Fig. 11** A comparison of output current ripple according to the load current ( $V_{in} = 36$  V,  $V_o = 50$  V).



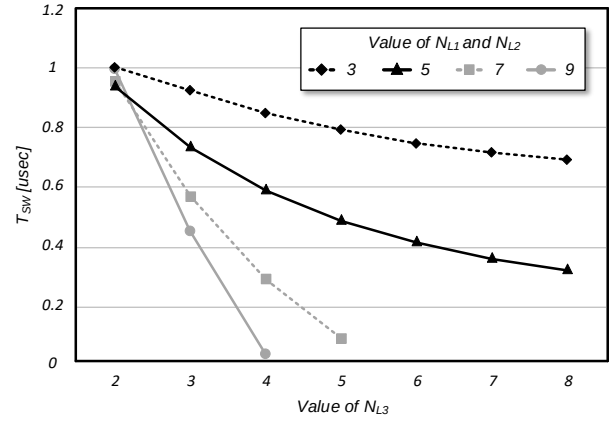
**Fig. 12** A comparison of output current ripple according to the input voltage ( $V_o = 50$  V,  $P_o = 750$  W).

cycle and commutation time were derived in A and B of Section III. Therefore, the slope of  $i_{Lk3}$  can be obtained for each mode of operation and the output current ripple can be determined accordingly. Moreover, the slope of  $i_{Lk3}$  is influenced by  $L_m$ ,  $L_{k1}$ ,  $L_{k2}$ ,  $L_{k3}$ , and  $L_{kt}$ .

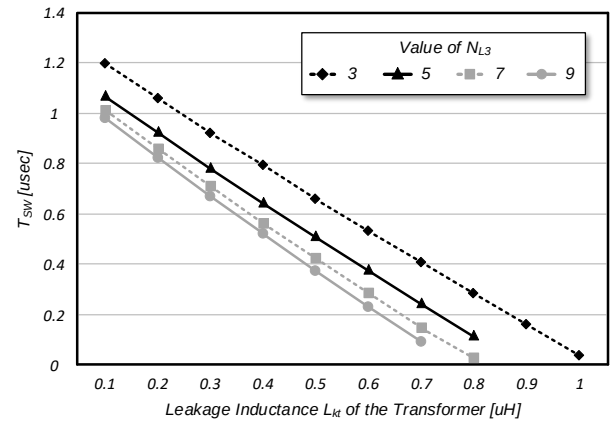
Fig. 9 presents the output current ripple according to the number of turns of the coupled inductors:  $N_{L1}$  ( $=N_{L2}$ ) and  $N_{L3}$ . Since the leakage inductance cannot be accurately predicted, assuming  $L_{k1}$  and  $L_{k2}$  are 10 % of  $L_m$ ,  $L_{k3}$  is 20 % of  $n_{L3}^2 L_m$  and  $L_{kt}$  is fixed at 0.3  $\mu$ H for the analysis. As shown in Fig. 9, for each  $N_{L1}$  ( $=N_{L2}$ ), there is  $N_{L3}$  where the output current ripple is minimized.

Fig. 10 shows the output current ripple according to  $N_{L3}$  and  $L_{kt}$  when  $N_{L1}$  ( $=N_{L2}$ ) is 4. As demonstrated in Fig. 10, a larger value of  $L_{kt}$  tends to further reduce the output current ripple.

Fig. 11 presents the output current ripples of the proposed and conventional converters according to the load current. For reference, the output current ripple of the conventional converter can be derived from Fig. 2. When comparing the output current ripple of both converters, the following assumptions are made:  $L_{kt}$  is 0.3  $\mu$ H,  $L_m$  is 2.512  $\mu$ H,  $L_{k1}$  and  $L_{k2}$  are 0.2512  $\mu$ H, and  $L_{k3}$  is 0.785  $\mu$ H. The selection of the inductance values for  $L_m$ ,  $L_{k1}$ ,  $L_{k2}$ , and  $L_{k3}$  will be discussed in detail in Section IV. As shown in Fig. 11, the output current



**Fig. 13** Soft-switching operation range according to  $N_{L1}$  ( $=N_{L2}$ ) and  $N_{L3}$  ( $V_{in} = 30$  V,  $V_o = 50$  V,  $I_o = 15$  A,  $L_{kt} = 0.3$   $\mu$ H).



**Fig. 14** Soft-switching operation range according to  $N_{L3}$  and  $L_{kt}$  ( $V_{in} = 30$  V,  $V_o = 50$  V,  $I_o = 15$  A,  $N_{L1} = N_{L2} = 3$ ).

ripple of the proposed converter was much smaller than that of the conventional converter over the entire load range. Moreover, as the load current increases, the output current ripple of the conventional converter exhibits a significant increase, whereas the proposed converter shows a considerably smaller increase in the output current ripple. As a result, the proposed converter is more advantageous than the conventional converter in terms of reducing the output voltage ripple due to the reduction in output current ripple, which would directly translate into a reduction in the size of the output capacitor.

Fig. 12 shows the output current ripples for the conventional and proposed converters according to the input voltage. The inductance values of  $L_m$ ,  $L_{k1}$ ,  $L_{k2}$ ,  $L_{k3}$ , and  $L_{kt}$  used to obtain the plot in Fig. 12 are as follows:  $L_{kt} = 0.3$   $\mu$ H,  $L_m = 2.512$   $\mu$ H,  $L_{k1}$  and  $L_{k2} = 0.2512$   $\mu$ H, and  $L_{k3} = 0.785$   $\mu$ H. These values are the same as those used in the previous analysis. Fig. 12 demonstrates that the output current ripple of the two converters is largest when the input voltage is within the middle range. Moreover, the output current ripple of the conventional converter is approximately four times larger than that of the proposed converter across all input voltage ranges.

As a result, if conventional and proposed converters maintain the same output voltage ripple, the proposed converter can reduce the number of output capacitors due to a significant

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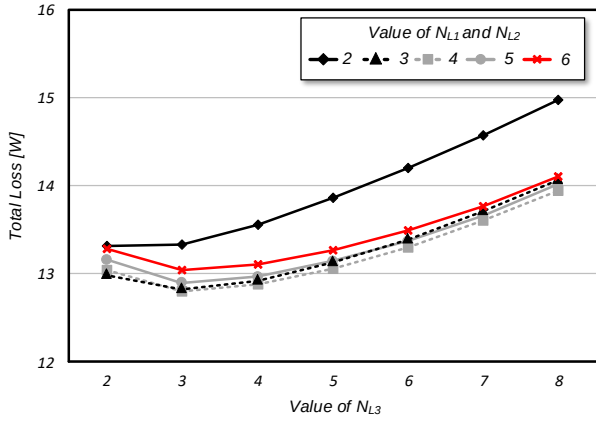


Fig. 15 The total converter loss according to  $N_{L1}$  ( $=N_{L2}$ ) and  $N_{L3}$ .

reduction in the output current ripple compared to the conventional Weinberg converter.

#### D. Leakage inductances for soft-switching operation

The proposed converter could achieve soft-switching operation for switches ( $M_1$  and  $M_2$ ) with the assistance of the leakage inductances in the circuit. However, if  $T_{CM(off)}$  shown in Fig. 6 is greater than  $(0.5-D)T_s$ , a hard-switching operation occurs as  $i_{Lkt}(=i_{ds1}=i_{ds2})$  does not drop to zero before the next switching operation. Therefore, the following conditions must be met to ensure soft-switching operation for  $M_1$  and  $M_2$ :

$$T_{sw} = (0.5 - D)T_s - T_{CM(off)} > 0. \quad (38)$$

Based on (32), (36), (37), and (38), the soft-switching operation conditions according to  $N_{L1}$  ( $=N_{L2}$ ) and  $N_{L3}$  are shown in Fig. 13. Furthermore, since the coupled inductor is implemented with a powder core, the inductance  $L_m$  is determined by the number of turns. Therefore, based on an experimental investigation of the coupled inductor implemented with various numbers of turns, we assume that  $L_{k1}$  and  $L_{k2}$  are 10 % of  $L_m$  and  $L_{k3}$  is 20 % of  $n_{L3}^2 L_m$  for the analysis. As shown in Fig. 13, when  $N_{L1}$  ( $=N_{L2}$ ) and  $N_{L3}$  decrease, the soft-switching operation becomes more favorable. In addition, when  $N_{L1}$  ( $=N_{L2}$ ) is 3, the soft-switching operation conditions according to the inductance of  $L_{kt}$  and the number of turns of  $N_{L3}$  are shown in Fig. 14. From Fig. 14, it is evident that a smaller  $L_{kt}$  results in more favorable soft-switching operation. Therefore, with smaller leakage inductances in the coupled inductor and transformer, soft-switching operation becomes easier.

#### IV. DESIGN OF OPTIMAL PARAMETERS

This section presents the optimal parameter selection based on loss analysis, including the analysis presented in Section III.

The efficiency of the proposed converter is primarily determined by the losses dissipated in switches, diodes, and magnetic components. Moreover, since soft-switching operation is possible, the turn-on loss of the switches can be ignored. The equations for analyzing the loss of each component are referenced in [42] and [43], and the coefficients

| material | temp.(°C) | $C_m$    | $x$   | $y$   |
|----------|-----------|----------|-------|-------|
| PL-7     | 60        | 1.93E-06 | 1.372 | 2.467 |
|          | 80        | 6.29E-07 | 1.411 | 2.619 |
|          | 100       | 2.21E-07 | 1.487 | 2.736 |

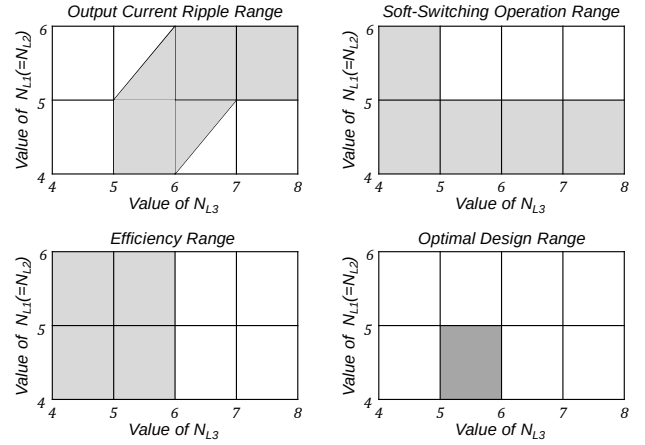


Fig. 16 The optimal design range of  $N_{L1}$  ( $=N_{L2}$ ) and  $N_{L3}$  considering the output current ripple, soft-switching operation, and efficiency

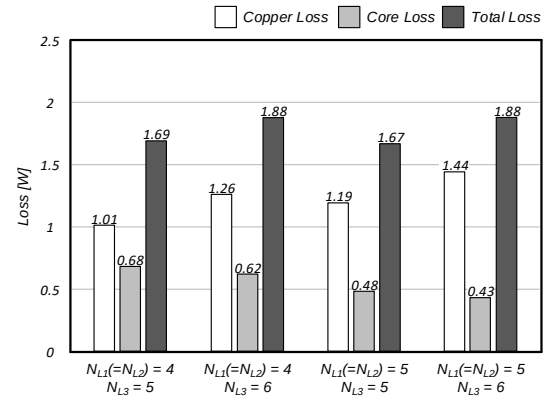


Fig. 17 The core and copper losses of the coupled inductor according to  $N_{L1}$  ( $=N_{L2}$ ) and  $N_{L3}$ .

for the transformer core loss are provided in Table III. The specifications and circuit parameters for the loss analysis are listed in Table IV in Section V. In particular, the input voltage is set to a minimum of 30 V, representing the worst-case scenario with the largest current.

Fig. 15 shows the total converter loss according to  $N_{L1}$  ( $=N_{L2}$ ) and  $N_{L3}$ , where the losses for each  $N_{L1}$  ( $=N_{L2}$ ) are very similar except for  $N_{L1}$  ( $=N_{L2}$ ) = 2. Therefore, selecting  $N_{L1}$  ( $=N_{L2}$ ) = 3 or higher is desirable in terms of total converter losses.

From the previous analysis, as shown in Fig. 13, it is evident that the soft-switching operation becomes easier as  $N_{L1}$  ( $=N_{L2}$ ) and  $N_{L3}$  decrease. However, as shown in Fig. 9, the output current ripple generally decreases as  $N_{L1}$  ( $=N_{L2}$ ) decreases and  $N_{L3}$  increases, except for  $N_{L1}$  ( $=N_{L2}$ ) = 3. In particular, the region indicated in the quadrilateral box in Fig. 9 corresponds to the range of  $N_{L1}$  ( $=N_{L2}$ ) and  $N_{L3}$  that exhibit a small output current ripple. Therefore, the number of turns of the coupled inductor should be selected in consideration of the region indicated in the quadrilateral box in terms of output current ripple.

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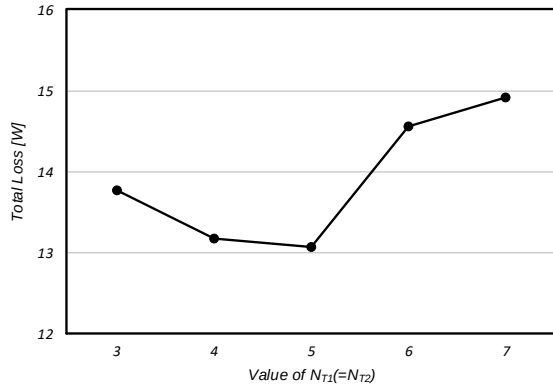


Fig. 18 The total converter loss according to variation of  $N_{T1}(=N_{T2})$ .

Ultimately, selecting the smallest possible  $N_{L1}(=N_{L2})$  and  $N_{L3}$  within the area indicated in the quadrilateral box in Fig. 9 (which demonstrates a small output current ripple) is also advantageous for ensuring the soft-switching operation of switches ( $M_1$  and  $M_2$ ) based on the results presented in Fig. 13. As mentioned previously, the number of turns of the coupled inductor should be selected appropriately considering the output current ripple, soft-switching operation, and the converter's efficiency. When considering the output current ripple, soft-switching operation, and total converter losses discussed in Figs. 9, 10, 13, 14, and 15, Fig. 16 presents the range of selectable  $N_{L1}(=N_{L2})$  and  $N_{L3}$ . More specifically, the range shown in Fig 16 satisfies the following conditions: the output current ripple is less than 7 A,  $T_{SW}$  is greater than 0.2  $\mu$ s to ensure soft-switching operation and total converter losses less than 15 W. Consequently, the number of turns for the coupled inductor that satisfies all conditions is  $N_{L1}(=N_{L2}) = 4$  and 5, and  $N_{L3} = 5$  and 6.

Fig. 17 presents the core and copper losses of the coupled inductor for each previously determined usable value of  $N_{L1}(=N_{L2})$  and  $N_{L3}$ . As demonstrated in Fig. 17, since the copper loss is larger than the core loss,  $N_{L1}(=N_{L2}) = 4$  and  $N_{L3} = 5$  are selected considering the heat balance for the core and wire of the coupled inductor and the margin for soft-switching operation.

Finally, to determine the optimal value of  $N_{T1}(=N_{T2})$ , the previously determined values of  $N_{L1}(=N_{L2}) = 4$  and  $N_{L3} = 5$  are utilized. Fig. 18 shows the total converter loss according to  $N_{T1}(=N_{T2})$ , where the total loss is lowest when  $N_{T1}(=N_{T2})$  is 5. In summary, the optimal values of  $N_{T1}(=N_{T2})$ ,  $N_{L1}(=N_{L2})$ , and  $N_{L3}$  are selected as 5, 4, and 5, respectively.

## V. EXPERIMENTAL RESULTS

To verify the validity of the proposed converter, this section presents the experimental results from prototypes of both the conventional and proposed converters based on the specifications in Table IV. Fig. 19 shows the control scheme of the proposed converter for experiments. As shown in Fig. 19, the proposed converter applies a type 2 controller to regulate the output voltage. Furthermore, the current flowing through switches  $M_1$  and  $M_2$  is monitored for peak current control. The prototypes of both converters

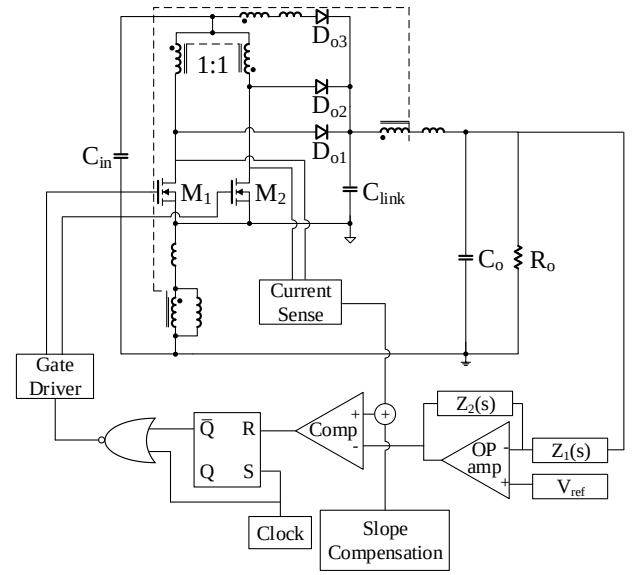


Fig. 19 Block diagram of the proposed converter control scheme.

TABLE IV  
PARAMETERS USED IN THE EXPERIMENT

| Parameter                            | Conventional converter                      | Proposed converter                    |
|--------------------------------------|---------------------------------------------|---------------------------------------|
| Input voltage $V_{in}$               | 30 V – 42 V                                 |                                       |
| Output voltage $V_o$                 | 50 V                                        |                                       |
| Rated power $P_o$                    | 750 W                                       |                                       |
| Switching frequency $f_{sw}$         | 100 kHz                                     |                                       |
| Transformer core                     | PQ2625                                      |                                       |
| Main switches                        | IPP045N10N3 (100 V/4.5 m $\Omega$ )         |                                       |
| Diodes                               | V60100C(100 V/ 0.66 V <sub>F</sub> / 260pF) |                                       |
| Number of turns                      | $N_{T1}(=N_{T2}) = 5$                       |                                       |
| leakage inductance $L_{kt}$          | 0.66 $\mu$ H                                |                                       |
| Coupled inductor core                | CH270125                                    |                                       |
| Number of turns                      | $N_{L1}(=N_{L2}) = 4$                       | $N_{L1}(=N_{L2}) = 4$<br>$N_{L3} = 5$ |
| magnetizing inductor $L_m$           | 2.73 $\mu$ H                                | 2.5 $\mu$ H                           |
| leakage inductance $L_{k1}(=L_{k2})$ | 0.29 $\mu$ H                                | 0.34 $\mu$ H                          |
| leakage inductance $L_{k3}$          | -                                           | 0.82 $\mu$ H                          |
| Output capacitor                     | MKT1820522015 (2.2 $\mu$ F)                 |                                       |
| Output capacitance $C_o$             | 24.2 $\mu$ F                                | 11 $\mu$ F                            |
| Link capacitor                       | RA4405K100-FA (4 $\mu$ F)                   |                                       |
| Link capacitance $C_{link}$          | -                                           | 8 $\mu$ F                             |

are depicted in Fig. 20, the experimental key waveforms of the two converters are shown in Fig. 21, and the voltage and current waveforms of switch  $M_1$  and diodes  $D_{o1}$  and  $D_{o3}$  are shown in Fig. 22. Furthermore, Fig. 23 represents the dynamic load experimental results of both the conventional and proposed converter.

First, Figs. 21(a) and (b) show the output current ripples of the two converters at an input voltage of 36 V, which represents the greatest output current ripple within the input voltage range. As shown in these figures, the output current ripples of the conventional converter were 15.4 A at 10 A load and 20.2 A at 15 A load. In comparison, the proposed converter exhibited significantly smaller current ripples of 5.5 A at 10 A load and 6.7 A at 15 A load. Hence, the proposed converter significantly reduced the output current ripple compared to the conventional converter. In addition, as the load current increased from 10 A to 15 A, the output current ripple of the conventional converter

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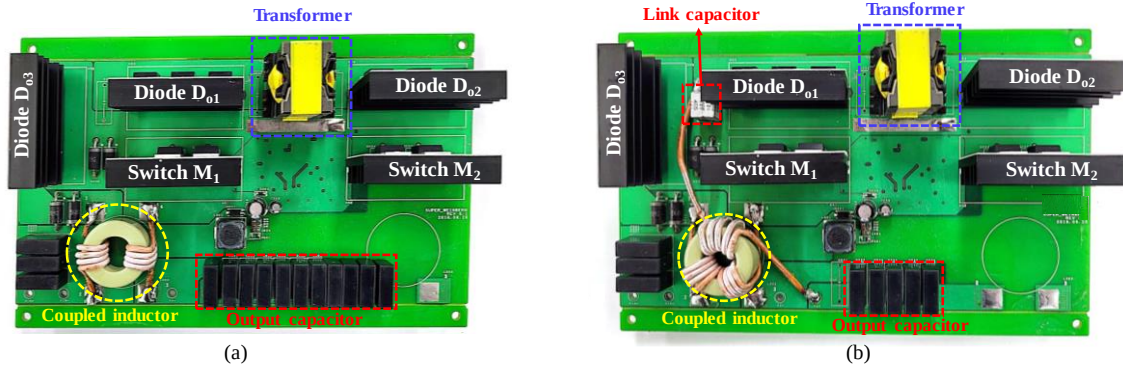


Fig. 20 Prototypes of (a) conventional converter and (b) proposed converter.

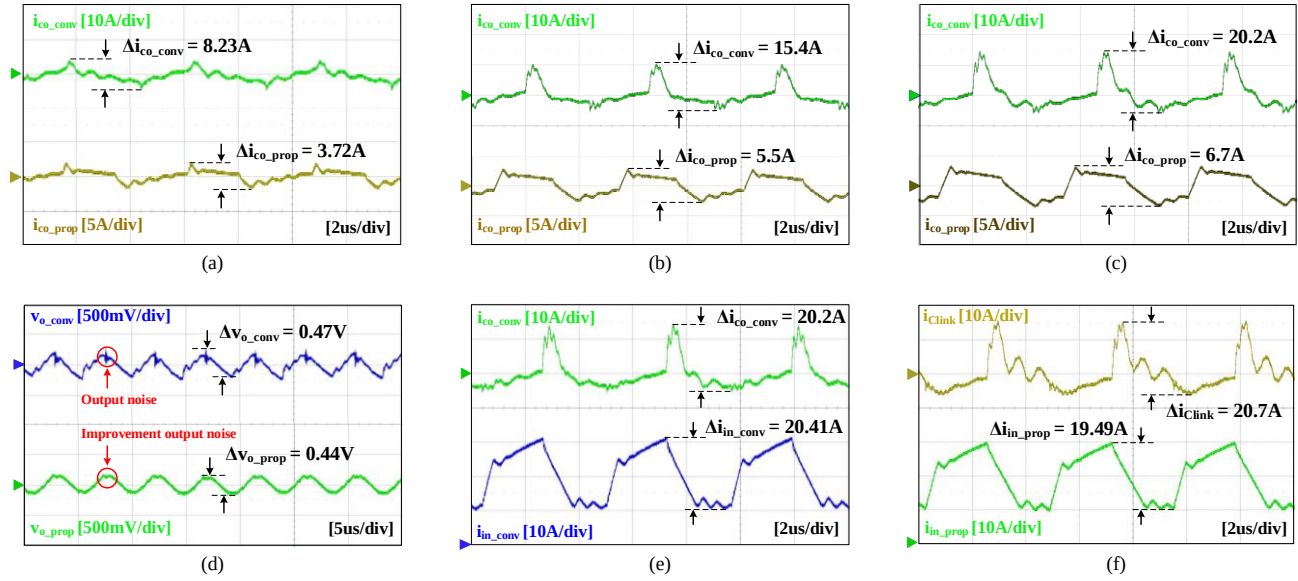


Fig. 21 Experimental key waveforms of proposed and conventional converter: (a) output capacitor current ripple at 3 A load, (b) output capacitor current ripple at 10 A load, (c) output capacitor current ripple at 15 A load (d) output voltage ripple at an input voltage of 36 V and 15 A load, and (e) output capacitor current and input current of the conventional converter at 15A load, (f) link capacitor current and input current of the proposed converter at 15A load.

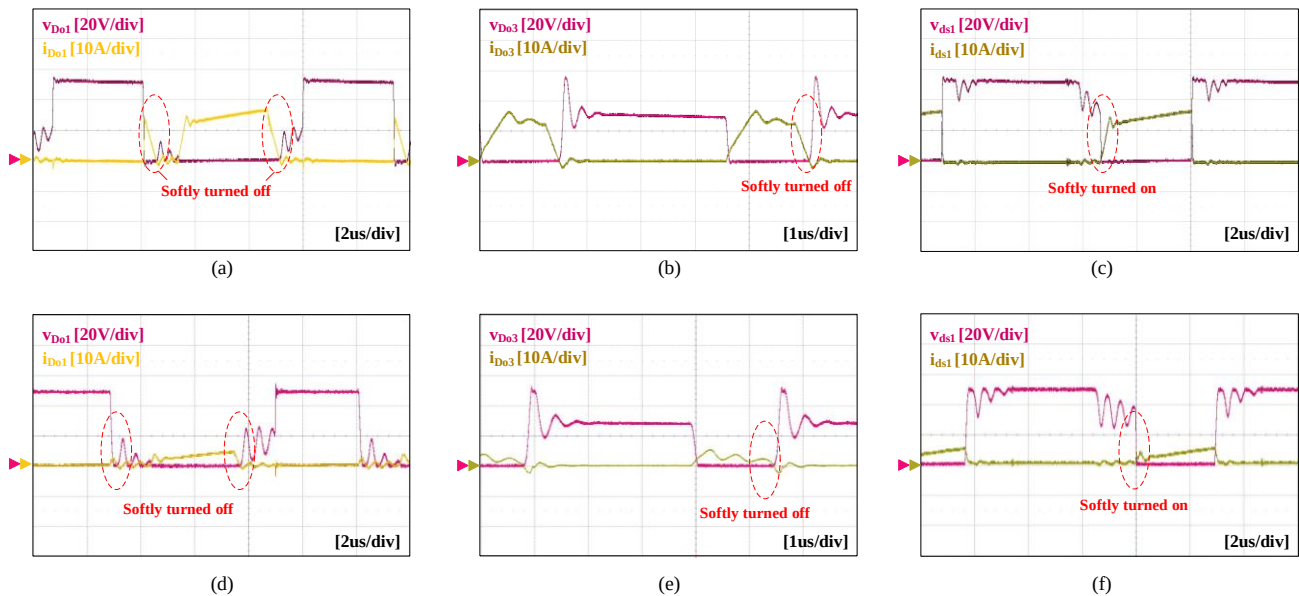
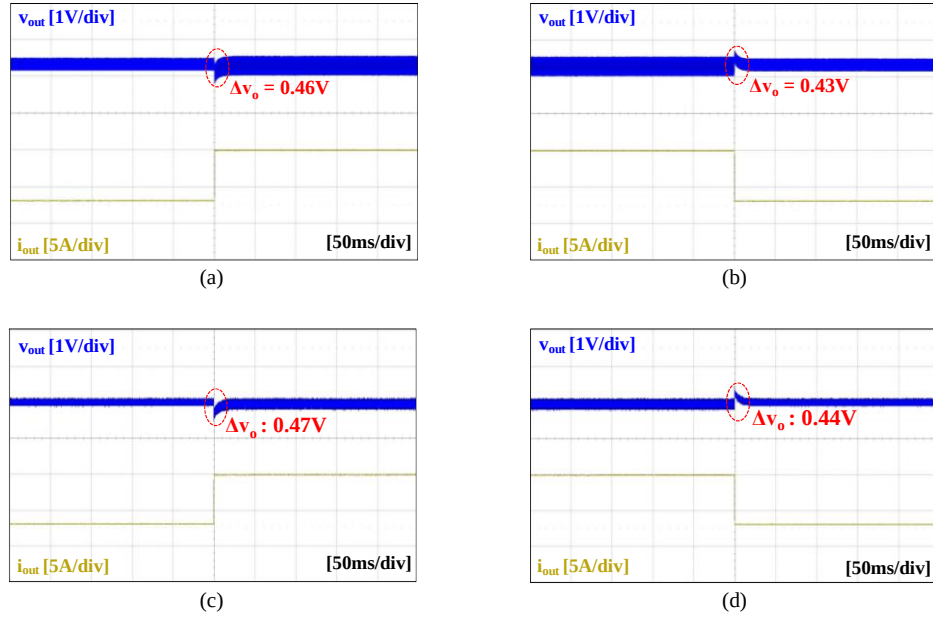
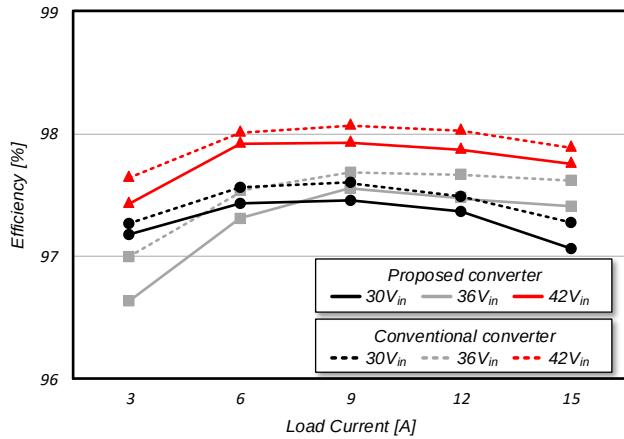


Fig. 22 Voltage and current waveforms of switch  $M_1$  and diodes  $D_{o1}$  and  $D_{o3}$  in proposed converter at input voltage of 30 V: (a) voltage and current of diode  $D_{o1}$  at 15 A load, (b) voltage and current of diode  $D_{o3}$  at 15 A load, (c) voltage and current of switch  $M_1$  at 15 A load, (d) voltage and current of diode  $D_{o1}$  at 3 A load, (e) voltage and current of diode  $D_{o3}$  at 3 A load and (f) voltage and current of switch  $M_1$  at 3 A load.

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**Fig. 23** Dynamic load test of conventional and proposed converters (a) 20 % load to 70 % load variation in conventional converter, (b) 70 % load to 20 % load variation in conventional converter, (c) 20 % load to 70 % load variation in proposed converter and (d) 70 % load to 20 % load variation in proposed converter



**Fig. 24** The measured efficiencies of the conventional and proposed converter according to load variation.

increased by 4.8 A. In contrast, the proposed converter exhibited a smaller increase of only 1.2 A. As depicted in Fig. 21(b), the rms current through the output capacitor of the conventional converter was 4.65 A<sub>rms</sub>. In comparison, the proposed converter demonstrated a significantly lower value of 2.02 A<sub>rms</sub>, which was approximately 2.3 times smaller than that of the conventional converter. In this paper, the proposed converter only utilizes film capacitors on the output side. However, in small satellites that utilize aluminum capacitors instead of film capacitors to reduce costs, the application of the proposed converter in this paper can be expected to increase the lifetime of the capacitors. This is due to the reduction in the rms current caused by the attenuation of the output current ripple [44], [45].

Fig. 21(d) presents the output voltage ripple of the proposed and conventional converters. As shown in Fig. 21(d), to maintain an output voltage ripple of approximately 0.44 V for both converters, the conventional converter needed to use

eleven output capacitors ( $C_o = 24.2 \mu\text{F}$ ), while the proposed converter needs to use only two link capacitors ( $C_{link} = 8 \mu\text{F}$ ) and five output capacitors ( $C_o = 11 \mu\text{F}$ ). Therefore, the number of capacitors required to satisfy an output voltage ripple of 0.44 V was reduced by approximately 36 % for the proposed converter compared to the conventional converter. Furthermore, as shown in Fig. 21(d), the proposed converter performs better in output voltage noise compared to the conventional converter. The improvement in output voltage noise in the proposed converter is due to the reduction of current ripple flowing through the output capacitors, as shown in Fig. 21 (a), (b), and (c). Although, as shown in Fig. 21(f), the current flowing through the link capacitor of the proposed converter also shows the significant ripple, resulting in overall noise levels that may be similar to the conventional converter. However, the proposed converter not only enhances output voltage noise compared to the conventional converter but also offers the advantage of reducing the number of output capacitors required to maintain the same output voltage ripple. Moreover, the total number of capacitors in the BDR for the EPS of a GEO satellite was further reduced due to the use of multiple BDRs in parallel for reliability and high power [46].

Figs. 22(a), (b), and (c) present the voltage and current waveforms of switch  $M_1$  and diodes  $D_{o1}$  and  $D_{o3}$  in the proposed converter under the worst case of soft-switching: minimum input voltage of 30 V and maximum load current of 15 A. As shown in Figs. 22(a) and (b),  $D_{o1}$  and  $D_{o3}$  did not have reverse recovery issues because the currents flowing through them gradually decreased to 0 A, after which they softly turned off. Furthermore, as shown in Fig. 22(c), since the current flowing through  $M_1$  gradually increased from 0 A during the turn-on transient, the voltage and current of  $M_1$  did not overlap significantly. Therefore,  $M_1$  could be turned on with soft-switching. Because  $D_{o2}$  and the  $M_2$  operated symmetrically to

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$M_1$  and  $D_{o1}$ ,  $M_2$  could also be turned on with soft-switching and  $D_{o2}$  did not experience any reverse recovery issues.

Fig. 23 shows the waveforms of the dynamic load test experimental results for the conventional and proposed converters. To conduct a dynamic load test, we increased the output capacitance of both converters to levels similar to those used in a real system. This was done because the previously presented output capacitance of both the conventional and proposed converters, as shown in Table IV, represents the minimum capacitance required to satisfy the condition of minimal output voltage ripple ( $\Delta v_o$  within 1 % of  $V_o$ ). The dynamic load specifications include a 20 % to 70 % load variation, and  $\Delta v_o$  should be within 1 %. In addition, the controller parameters and slew rate for the dynamic load tests of both the conventional and proposed converters were conducted under the same conditions. As shown in Fig. 23, when comparing the output voltage variation of the conventional and proposed converters at a similar level, it is observed that the conventional converter has an output capacitance of 368  $\mu\text{F}$ , whereas the proposed converter has a total capacitance of 258  $\mu\text{F}$ . This includes a link capacitance of 11  $\mu\text{F}$  and an output capacitance of 247  $\mu\text{F}$ . Therefore, based on the same dynamic load performance, the proposed converter can reduce the total capacitance by 110  $\mu\text{F}$  compared to the conventional converter.

Finally, Fig. 24 presents the efficiencies of the conventional and proposed converters, which were measured according to load current variations at input voltages of 30 V, 36 V, and 42 V. As shown in Fig. 24, the efficiencies of the two converters were similar across the entire load range. Furthermore, the average efficiency of the proposed converter (approximately 97.5%) was not significantly different from the average efficiency of the conventional converter (approximately 97.6%).

## VI. CONCLUSION

This paper presented a converter that can improve the drawbacks of the conventional Weinberg converter by adding a few link capacitors and auxiliary windings of a coupled inductor. The proposed converter significantly reduced the output current ripple compared to the conventional converter due to the leakage inductance of the auxiliary windings. Although the proposed converter required a few link capacitors, their size, and capacitance were not substantial. Furthermore, the number of output capacitors could be significantly reduced in the proposed converter due to the small output current ripple. **Another method to effectively reduce output current ripple is by adding a post LC filter, but it has the drawback of requiring magnetic devices capable of handling large load currents. On the other hand, the proposed converter has the advantage of reducing output current ripple using only auxiliary windings, without the need for additional magnetic devices.** Therefore, the overall volume and weight of the proposed converter could be reduced, even with the added link capacitors. In addition, the proposed converter had a substantially lower rms current flowing through the output capacitor compared to the

conventional converter. Consequently, if both converters used the same output capacitance, the proposed converter would reduce the stress on the output capacitor. Moreover, as the load current increased, the output current ripple of the conventional converter significantly increased, while the proposed converter exhibited a smaller increase in the output current ripple. Hence, the proposed converter would be more suitable for high-power applications compared to the conventional converter.

Despite the addition of auxiliary windings in the proposed converter compared to the conventional converter, it achieved a similar efficiency level of approximately 97.5 % across the entire load range. This high efficiency of the proposed converter was achieved through a reduction in losses due to three factors: the soft-switching operation, the elimination of the reverse recovery problem of diodes, and optimal parameter design through loss analysis. Moreover, the proposed converter reduced the overall weight of the conventional converter from 404 to 393 g by reducing the number of output capacitors, which was attributable to the reduction in output current ripple. Furthermore, the weight reduction effect of the proposed converter would be more significant as multiple BDR modules are typically used in parallel to enhance reliability, redundancy, and power. Based on these results, the proposed converter is expected to be suitable for high-power applications and achieve miniaturization and weight reduction of EPS for GEO satellites.

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# Reply to Reviewers' Comments

- **Manuscript ID: TPEL-Reg-2023-07-1723.R2**
- **Manuscript Title: Low Output Current Ripple Weinberg Converter for Battery Discharge Regulator in Geostationary Satellite**

We wish to thank the Editor-in-Chief, Co-Editor-in-Chief, Associate Editor, and all reviewers for their time and valuable comments on our paper. We have made necessary changes required by the reviewers. As a result, the revised manuscript will be much more valuable paper. In the revised manuscript, the **blue color** is used for the changed and added parts. We would like to explain our responses as follows:

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▪ **Comment:**

The reviewer(s) have conditionally recommended publication provided you adequately respond to their concerns and incorporate the major revisions they requested. Please note, this is only a conditional acceptance. If the editors and reviewers do not believe the changes to your manuscript adequately respond to their concerns, there is still a chance the paper may be declined for publication. Therefore, I invite you to respond to the reviewer(s)' comments and revise your manuscript.

→ **Answer:** Thank you for your valuable decision. Additionally, we would like to express our gratitude to all the reviewers who have contributed to reviewing our paper, helping to enhance its quality. We will carefully consider and sincerely respond to the thoughtful comment provided by the reviewers.

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**Reply to Co-EIC 1’s comments:**

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▪ **Comment:**

Please highlight the contribution and the difference with the paper noted in one of the reviewer's comments. Although the authors have answered the questions and made enhancement to the paper, there is still some comments that needs to be addressed.

→ **Answer:** Thank you for your valuable comment. As you suggested, we will respond diligently to the comments from the reviewers. Additionally, we will highlight the differences and contributions with the paper mentioned in one of the reviewer's comments.

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## Reply to Associate Editor 2's comments:

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▪ **Comment:**

Good contribution from authors and reviewers. Please address the following comments to advance to next stages.

- address the comments from the reviewers,
- further bring out the novelty and contributions especially around inductors and commercial impact,
- thoroughly consider the published work and compare your work,

Post LC filter and implementation of the required inductor integrated with other existing inductor should be clearly mentioned in the paper text, this should be discussed align with key results.

➔ **Answer:** Thank you for your valuable comments. We tried to do our best to revise the paper in accordance with the reviewer's comments. In this revision, we have taken into account the comments of the reviewers and made additions and changes to the content of the paper. In addition, we have provided responses to the request you made on the following pages.

- **Post LC filter and implementation of the required inductor integrated with other existing inductor should be clearly mentioned in the paper text, this should be discussed align with key results.**

Thank you very much for your kind and detailed review. Adding a post LC filter to the conventional converter is highly effective in reducing output voltage ripple. However, to add an LC filter, additional magnetic devices capable of handling high load currents need to be included. On the other hand, the proposed converter achieves the benefits of adding an LC filter without the need for additional magnetic devices by incorporating auxiliary windings into the coupled inductor. Therefore, the proposed converter is expected to increase power density by reducing output voltage ripple without the need for additional magnetic devices.

Based on your opinion, we have modified the paper by adding a sentence, as shown below, detailing the differences between using the post-LC filter and the proposed converter.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <b>→ [In revised manuscript]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><b>(Previous part in INTRODUCTION)</b></p> <p>Table I shows the comparison results between several converters with continuous input and output currents and the proposed converter, under the same specifications (<math>V_{in} = 36\text{ V}</math>, <math>V_o = 50\text{ V}</math>, <math>P_o = 750\text{ W}</math>, <math>f_{sw} = 100\text{ kHz}</math>). The inductance is set to <math>10\text{ }\mu\text{H}</math>, and the output capacitance is set to <math>15.4\text{ }\mu\text{F}</math> for the comparison. As shown in Table I, both the super boost converter and Cuk converter have the disadvantage of operating with hard switching and facing the issue of diode reverse recovery, despite the advantage of using fewer semiconductor components. Moreover, the large peak current flowing through each device poses limitations for use in high-current applications. In contrast, both the conventional Weinberg converter and the proposed converter, although using a larger number of semiconductor devices compared to the two converters mentioned earlier, have the advantage of being capable of soft switching operation and avoiding diode reverse recovery issues. Additionally, the peak current flowing through the semiconductor devices is relatively small, making them advantageous for high-current applications. In particular, the proposed converter has a smaller output current ripple compared to other converters, resulting in a smaller output voltage ripple, which can provide more</p> |

precise voltage to the load. Therefore, it can be observed from Table I that the proposed converter among the topologies presented is the most suitable for satellite applications.

Furthermore, the converter proposed in [46] to reduce the output current ripple of the conventional Weinberg converter and the converter proposed in this paper generally share a similar topology. However, the two converters differ in the number of magnetic devices used to improve the output current ripple and in the methods employed for optimizing the converter design.

### (Revised part in INTRODUCTION)

Table I shows the comparison results between several converters with continuous input and output currents and the proposed converter, under the same specifications ( $V_{in} = 36$  V,  $V_o = 50$  V,  $P_o = 750$  W,  $f_{sw} = 100$  kHz). The inductance is set to 10  $\mu$ H, and the output capacitance is set to 15.4  $\mu$ F for the comparison. As shown in Table I, both the super boost converter and Cuk converter have the disadvantage of operating with hard switching and facing the issue of diode reverse recovery, despite the advantage of using fewer semiconductor components. Moreover, the large peak current flowing through each device poses limitations for use in high-current applications. In contrast, both the conventional Weinberg converter and the proposed converter, although using a larger number of semiconductor devices compared to the two converters mentioned earlier, have the advantage of being capable of soft switching operation and avoiding diode reverse recovery issues. Additionally, the peak current flowing through the semiconductor devices is relatively small, making them advantageous for high-current applications. In particular, the proposed converter has a smaller output current ripple compared to other converters, resulting in a smaller output voltage ripple, which can provide more precise voltage to the load. Therefore, it can be observed from Table I that the proposed converter among the topologies presented is the most suitable for satellite applications.

On the other hand, one effective method to significantly reduce the output voltage ripple of the conventional Weinberg converter is to add a post LC filter to the output side. However, to add an LC

filter, additional magnetic devices capable of handling high load currents need to be included. Moreover, due to the continuous increase in load current with the ongoing rise in satellite power, the burden on the magnetic components constituting the LC filter will also increase. In contrast, the proposed converter achieves the benefits of adding an LC filter without the need for additional magnetic devices by adding auxiliary windings to the coupled inductor. Therefore, the proposed converter has a significant advantage in reducing the output current ripple without requiring additional magnetic devices to handle large load currents.

Furthermore, the converter proposed in [41] to reduce the output current ripple of the conventional Weinberg converter and the converter proposed in this paper generally share a similar topology. However, the two converters differ in the number of magnetic devices used to improve the output current ripple and in the methods employed for optimizing the converter design.

**(Previous part in CONCLUSION)**

This paper presented a converter that can improve the drawbacks of the conventional Weinberg converter by adding a few link capacitors and auxiliary windings of a coupled inductor. The proposed converter significantly reduced the output current ripple compared to the conventional converter due to the leakage inductance of the auxiliary windings. Although the proposed converter required a few link capacitors, their size, and capacitance were not substantial. Furthermore, the number of output capacitors could be significantly reduced in the proposed converter due to the small output current ripple. Therefore, the overall volume and weight of the proposed converter could be reduced, even with the added link capacitors. In addition, the proposed converter had a substantially lower rms current flowing through the output capacitor compared to the conventional converter. Consequently, if both converters used the same output capacitance, the proposed converter would reduce the stress on the output capacitor. Moreover, as the load current increased, the output current ripple of the conventional

converter significantly increased, while the proposed converter exhibited a smaller increase in the output current ripple. Hence, the proposed converter would be more suitable for high-power applications compared to the conventional converter.

### **(Revised part in CONCLUSION)**

This paper presented a converter that can improve the drawbacks of the conventional Weinberg converter by adding a few link capacitors and auxiliary windings of a coupled inductor. The proposed converter significantly reduced the output current ripple compared to the conventional converter due to the leakage inductance of the auxiliary windings. Although the proposed converter required a few link capacitors, their size, and capacitance were not substantial. Furthermore, the number of output capacitors could be significantly reduced in the proposed converter due to the small output current ripple. Another method to effectively reduce output current ripple is by adding a post LC filter, but it has the drawback of requiring magnetic devices capable of handling large load currents. On the other hand, the proposed converter has the advantage of reducing output current ripple using only auxiliary windings, without the need for additional magnetic devices. Therefore, the overall volume and weight of the proposed converter could be reduced, even with the added link capacitors. In addition, the proposed converter had a substantially lower rms current flowing through the output capacitor compared to the conventional converter. Consequently, if both converters used the same output capacitance, the proposed converter would reduce the stress on the output capacitor. Moreover, as the load current increased, the output current ripple of the conventional converter significantly increased, while the proposed converter exhibited a smaller increase in the output current ripple. Hence, the proposed converter would be more suitable for high-power applications compared to the conventional converter.

Reply to Reviewer 3’s comments:

▪ Comment:

Thanks for revising the paper. The point that the difference is about post LC filter and implementation of the required inductor integrated with other existing inductor should be clearly mentioned in the paper text. Also pulsating current exist in clink and noise due to pulsating ground current should be discussed and measured and compared.

➔ **Answer:** Thank you for your valuable comments. Adding a post LC filter to the conventional converter is highly effective in reducing output voltage ripple. However, to add an LC filter, additional magnetic devices capable of handling high load currents need to be included. On the other hand, the proposed converter achieves the benefits of adding an LC filter without the need for additional magnetic devices by incorporating auxiliary windings into the coupled inductor. Therefore, the proposed converter is expected to increase power density by reducing output voltage ripple without the need for additional magnetic devices.

Based on your opinion, we have modified the paper by adding a sentence, as shown below, detailing the differences between using the post-LC filter and the proposed converter.

➔ [In revised manuscript]

(Previous part in INTRODUCTION)

Table I shows the comparison results between several converters with continuous input and output currents and the proposed converter, under the same specifications ( $V_{in} = 36\text{ V}$ ,  $V_o = 50\text{ V}$ ,  $P_o = 750\text{ W}$ ,  $f_{sw} = 100\text{ kHz}$ ). The inductance is set to  $10\text{ }\mu\text{H}$ , and the output capacitance is set to  $15.4\text{ }\mu\text{F}$  for the comparison. As shown in Table I, both the super boost converter and Cuk converter have the disadvantage of operating with hard switching and facing the issue of diode reverse recovery, despite

the advantage of using fewer semiconductor components. Moreover, the large peak current flowing through each device poses limitations for use in high-current applications. In contrast, both the conventional Weinberg converter and the proposed converter, although using a larger number of semiconductor devices compared to the two converters mentioned earlier, have the advantage of being capable of soft switching operation and avoiding diode reverse recovery issues. Additionally, the peak current flowing through the semiconductor devices is relatively small, making them advantageous for high-current applications. In particular, the proposed converter has a smaller output current ripple compared to other converters, resulting in a smaller output voltage ripple, which can provide more precise voltage to the load. Therefore, it can be observed from Table I that the proposed converter among the topologies presented is the most suitable for satellite applications.

Furthermore, the converter proposed in [46] to reduce the output current ripple of the conventional Weinberg converter and the converter proposed in this paper generally share a similar topology. However, the two converters differ in the number of magnetic devices used to improve the output current ripple and in the methods employed for optimizing the converter design.

### **(Revised part in INTRODUCTION)**

Table I shows the comparison results between several converters with continuous input and output currents and the proposed converter, under the same specifications ( $V_{in} = 36$  V,  $V_o = 50$  V,  $P_o = 750$  W,  $f_{sw} = 100$  kHz). The inductance is set to 10  $\mu$ H, and the output capacitance is set to 15.4  $\mu$ F for the comparison. As shown in Table I, both the super boost converter and Cuk converter have the disadvantage of operating with hard switching and facing the issue of diode reverse recovery, despite the advantage of using fewer semiconductor components. Moreover, the large peak current flowing through each device poses limitations for use in high-current applications. In contrast, both the conventional Weinberg converter and the proposed converter, although using a larger number of semiconductor devices compared to the two converters mentioned earlier, have the advantage of being

capable of soft switching operation and avoiding diode reverse recovery issues. Additionally, the peak current flowing through the semiconductor devices is relatively small, making them advantageous for high-current applications. In particular, the proposed converter has a smaller output current ripple compared to other converters, resulting in a smaller output voltage ripple, which can provide more precise voltage to the load. Therefore, it can be observed from Table I that the proposed converter among the topologies presented is the most suitable for satellite applications.

On the other hand, one effective method to significantly reduce the output voltage ripple of the conventional Weinberg converter is to add a post LC filter to the output side. However, to add an LC filter, additional magnetic devices capable of handling high load currents need to be included. Moreover, due to the continuous increase in load current with the ongoing rise in satellite power, the burden on the magnetic components constituting the LC filter will also increase. In contrast, the proposed converter achieves the benefits of adding an LC filter without the need for additional magnetic devices by adding auxiliary windings to the coupled inductor. Therefore, the proposed converter has a significant advantage in reducing the output current ripple without requiring additional magnetic devices to handle large load currents.

Furthermore, the converter proposed in [41] to reduce the output current ripple of the conventional Weinberg converter and the converter proposed in this paper generally share a similar topology. However, the two converters differ in the number of magnetic devices used to improve the output current ripple and in the methods employed for optimizing the converter design.

### **(Previous part in CONCLUSION)**

This paper presented a converter that can improve the drawbacks of the conventional Weinberg converter by adding a few link capacitors and auxiliary windings of a coupled inductor. The proposed converter significantly reduced the output current ripple compared to the conventional converter due to

the leakage inductance of the auxiliary windings. Although the proposed converter required a few link capacitors, their size, and capacitance were not substantial. Furthermore, the number of output capacitors could be significantly reduced in the proposed converter due to the small output current ripple. Therefore, the overall volume and weight of the proposed converter could be reduced, even with the added link capacitors. In addition, the proposed converter had a substantially lower rms current flowing through the output capacitor compared to the conventional converter. Consequently, if both converters used the same output capacitance, the proposed converter would reduce the stress on the output capacitor. Moreover, as the load current increased, the output current ripple of the conventional converter significantly increased, while the proposed converter exhibited a smaller increase in the output current ripple. Hence, the proposed converter would be more suitable for high-power applications compared to the conventional converter.

### (Revised part in CONCLUSION)

This paper presented a converter that can improve the drawbacks of the conventional Weinberg converter by adding a few link capacitors and auxiliary windings of a coupled inductor. The proposed converter significantly reduced the output current ripple compared to the conventional converter due to the leakage inductance of the auxiliary windings. Although the proposed converter required a few link capacitors, their size, and capacitance were not substantial. Furthermore, the number of output capacitors could be significantly reduced in the proposed converter due to the small output current ripple. Another method to effectively reduce output current ripple is by adding a post LC filter, but it has the drawback of requiring magnetic devices capable of handling large load currents. On the other hand, the proposed converter has the advantage of reducing output current ripple using only auxiliary windings, without the need for additional magnetic devices. Therefore, the overall volume and weight of the proposed converter could be reduced, even with the added link capacitors. In addition, the

proposed converter had a substantially lower rms current flowing through the output capacitor compared to the conventional converter. Consequently, if both converters used the same output capacitance, the proposed converter would reduce the stress on the output capacitor. Moreover, as the load current increased, the output current ripple of the conventional converter significantly increased, while the proposed converter exhibited a smaller increase in the output current ripple. Hence, the proposed converter would be more suitable for high-power applications compared to the conventional converter.

Furthermore, if the pulsating ground current in your comment corresponds to the output current ripple of the conventional converter, the comparison with the pulsating current flowing through  $C_{link}$  is shown in the Fig. R3.1 below. Since the output current ripple of the conventional converter and the current ripple through  $C_{link}$  in the proposed converter are at similar levels, it is expected that the noise levels will be comparable. However, while the output current ripple of the conventional converter is directly associated with bus voltage ripple, the proposed converter, thanks to the auxiliary winding of the coupled inductor, does not significantly impact bus voltage ripple even when there is a substantial ripple current flowing through  $C_{link}$ . The bus filter supplies power to all electrical units of the satellite. In other words, the conventional converter's output ripple and noise have the disadvantage of degrading the quality of power delivered to all electrical units. Therefore, considering the importance of bus voltage stability in satellite systems, the proposed converter, which effectively reduces bus voltage noise, can be considered superior for satellite systems.

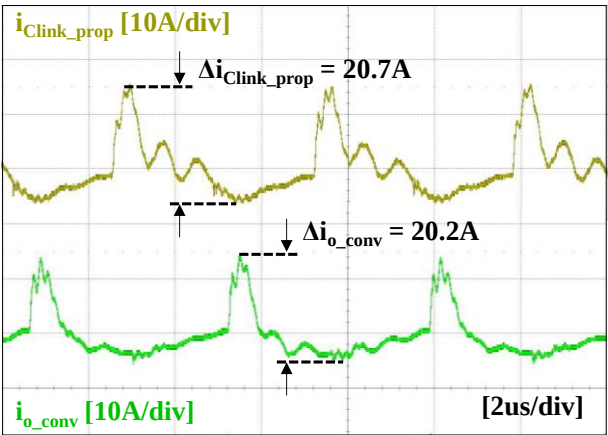


Fig. R3. 1 The ripple currents through the circuit components (a) Link capacitor in proposed converter, (b) Output capacitor in conventional converter.

1        Additionally, information about the output current ripple of the conventional converter and the current ripple  
2        flowing through  $C_{link}$  in the proposed converter is discussed in the third paragraph of Section V of this paper.  
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4        Further details can be found below.  
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7        Although, as shown in Fig. 21(f), the current flowing through the link capacitor of the proposed converter  
8        also shows the significant ripple, resulting in overall noise levels that may be similar to the conventional  
9        converter. However, the proposed converter not only enhances output voltage noise compared to the  
10        conventional converter but also offers the advantage of reducing the number of output capacitors required to  
11        maintain the same output voltage ripple.  
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**Reply to Reviewer 5's comments:**

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▪ **Comment:**

In this paper, a Weinberg converter with low output current ripple is proposed, where an inductor is added in the output. However, the control of the converter is essential. The detailed comments are listed in the following.

1. In the comparison part, the converter in [41] should be added. The number in the proposed Weinberg converter is increased, so the comparison about efficiency, complexity and power density should be evaluated.
2. In the conventional Weinberg converter, increasing the output capacitance may also reduce the output voltage ripple. What is the advantage of the proposed converter? Maybe you can added the capacitance reduction of the proposed converter in Section III.
3. Control scheme of the proposed converter is essential.

→ **Answer:** Thank you for your valuable comments. We did our best to revise our manuscript according to your comments. We would like to explain our detail responses in next page

▪ **Comment 1:**

In the comparison part, the converter in [41] should be added. The number in the proposed Weinberg converter is increased, so the comparison about efficiency, complexity and power density should be evaluated.

→ **Answer:** Thank you for your insightful comment. We have added reference [41] into Table I, as per your feedback, and included a comparison of topologies. The proposed converter has the advantage of significantly lower peak current for switches and diodes compared to the converters mentioned in references [39] and [40], while also enabling soft switching. Additionally, while the proposed converter has the same total number of components (diodes, switches, magnetic devices, and capacitors) as the converter in [36], it offers significant advantages such as very small output current ripple and output voltage ripple. Furthermore, the proposed converter has the advantage of reducing the number of magnetic devices compared to the converter mentioned in [41].

According to your opinion, the content of the topology comparison, including the referenced paper [41], has been added as shown below.

→ [In revised manuscript]

(Previous TABLE I)

TABLE I  
COMPARISON OF TOPOLOGIES

|                             | Weinberg<br>converter<br>[36] | Super boost<br>converter<br>[39] | Cuk<br>converter<br>[40] | Proposed<br>converter |
|-----------------------------|-------------------------------|----------------------------------|--------------------------|-----------------------|
| Number of the diode         | 3                             | 1                                | 1                        | 3                     |
| Number of the switch        | 2                             | 1                                | 1                        | 2                     |
| Number of the magnetic core | 2                             | 2                                | 2                        | 2                     |
| Number of the capacitor     | 2                             | 2                                | 2                        | 2                     |
| Soft-switching operation    | O                             | X                                | X                        | O                     |
| Diode reverse recovery      | X                             | O                                | O                        | X                     |
| Peak current of the switch  | 14.1A                         | 30.91A                           | 56.8A                    | 15.4A                 |
| Peak current of the diode   | 14.1A                         | 30.91A                           | 56.8A                    | 15.4A                 |
| Output current ripple       | 15.5A                         | 10.2A                            | 21.2A                    | 6.23A                 |
| Output voltage ripple       | 0.41V                         | 0.83V                            | 1.72V                    | 0.29V                 |

(Revised TABLE I)

| TABLE I<br>COMPARISON OF TOPOLOGIES |       |        |       |       |                       |
|-------------------------------------|-------|--------|-------|-------|-----------------------|
|                                     | [36]  | [39]   | [40]  | [41]  | Proposed<br>converter |
| Number of the diode                 | 3     | 1      | 1     | 3     | 3                     |
| Number of the switch                | 2     | 1      | 1     | 2     | 2                     |
| Number of the magnetic core         | 2     | 2      | 2     | 3     | 2                     |
| Number of the capacitor             | 2     | 2      | 2     | 2     | 2                     |
| Soft-switching operation            | O     | X      | X     | O     | O                     |
| Diode reverse recovery              | X     | O      | O     | X     | X                     |
| Peak current of the switch          | 14.1A | 30.91A | 56.8A | 15A   | 15.4A                 |
| Peak current of the diode           | 14.1A | 30.91A | 56.8A | 15A   | 15.4A                 |
| Output current ripple               | 15.5A | 10.2A  | 21.2A | 4.26A | 6.23A                 |
| Output voltage ripple               | 0.41V | 0.83V  | 1.72V | 0.17V | 0.29V                 |

→ [In revised manuscript]

(Previous part in INTRODUCTION)

Table I shows the comparison results between several converters with continuous input and output currents and the proposed converter, under the same specifications ( $V_{in} = 36\text{ V}$ ,  $V_o = 50\text{ V}$ ,  $P_o = 750\text{ W}$ ,  $f_{sw} = 100\text{ kHz}$ ). The inductance is set to  $10\text{ }\mu\text{H}$ , and the output capacitance is set to  $15.4\text{ }\mu\text{F}$  for the comparison. As shown in Table I, both the super boost converter and Cuk converter have the disadvantage of operating with hard switching and facing the issue of diode reverse recovery, despite the advantage of using fewer semiconductor components. Moreover, the large peak current flowing through each device poses limitations for use in high-current applications. In contrast, both the conventional Weinberg converter and the proposed converter, although using a larger number of semiconductor devices compared to the two converters mentioned earlier, have the advantage of being capable of soft switching operation and avoiding diode reverse recovery issues. Additionally, the peak current flowing through the semiconductor devices is relatively small, making them advantageous for high-current applications. In particular, the proposed converter has a smaller output current ripple compared to other converters, resulting in a smaller output voltage ripple, which can provide more precise voltage to the load. Therefore, it can be observed from Table I that the proposed converter among the topologies presented is the most suitable for satellite applications.

Furthermore, the converter proposed in [41] to reduce the output current ripple of the conventional Weinberg converter and the converter proposed in this paper generally share a similar topology. However, the two converters differ in the number of magnetic devices used to improve the output current ripple and in the methods employed for optimizing the converter design.

### (Revised part in INTRODUCTION)

Table I shows the comparison results between several converters with continuous input and output currents and the proposed converter, under the same specifications ( $V_{in} = 36$  V,  $V_o = 50$  V,  $P_o = 750$  W,  $f_{sw} = 100$  kHz). The inductance is set to 10  $\mu$ H, and the output capacitance is set to 15.4  $\mu$ F for the comparison. As shown in Table I, [39] and [40] have the disadvantage of operating with hard switching and facing the issue of diode reverse recovery, despite the advantage of using fewer semiconductor components. Moreover, the large peak current flowing through each device poses limitations for use in high-current applications. In contrast, [36], [41], and the proposed converter, although using a larger number of semiconductor devices compared to the two previously mentioned converters, have the advantage of being capable of soft switching operation and avoiding diode reverse recovery issues. Additionally, the peak current flowing through the semiconductor devices is relatively small, making them advantageous for high-current applications. However, [36] has a drawback that the output voltage ripple is larger compared to [41] and the proposed converter due to the larger output current ripple. The proposed converter, as well as the one mentioned in reference [41], exhibits significantly lower output voltage and current ripples compared to other converters. Although the proposed converter and [41] share a similar topology to reduce output current ripple, the proposed converter achieves this with one less magnetic device compared to [41]. Additionally, the methods employed for optimizing the converter design are also different. Furthermore, since BDRs are generally used in parallel with multiple converters, these differences in weight and volume become even greater. Therefore, it can be

observed from Table I that the proposed converter among the topologies presented is the most suitable for satellite applications.

~~Furthermore, the converter proposed in [41] to reduce the output current ripple of the conventional Weinberg converter and the converter proposed in this paper generally share a similar topology. However, the two converters differ in the number of magnetic devices used to improve the output current ripple and in the methods employed for optimizing the converter design.~~

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▪ **Comment 2:**

In the conventional Weinberg converter, increasing the output capacitance may also reduce the output voltage ripple. What is the advantage of the proposed converter? Maybe you can added the capacitance reduction of the proposed converter in Section III.

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→ **Answer:** Thank you for your helpful comment.

Your comment is accurate. However, the main focus of our paper is to reduce the output current ripple in the conventional Weinberg converter, which would allow for a reduction in the required number of output capacitors while maintaining the same bus voltage ripple. In response to your feedback, we have included information in Section III about the decrease in the number of output capacitors due to the reduction in output current ripple, as shown below.

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→ [In revised manuscript]

**(Previous part in ANALYSIS OF THE PROPOSED CONVERTER)**

*C. Output current ripple*

As shown in Fig. 3, the output current of the proposed converter is equal to  $i_{Lk3}$ . Furthermore, the voltage applied to  $L_{k3}$  for each mode was derived in Section II, and the effective duty cycle and commutation time were derived in A and B of Section III. Therefore, the slope of  $i_{Lk3}$  can be obtained for each mode of operation and the output current ripple can be determined accordingly. Moreover, the slope of  $i_{Lk3}$  is influenced by  $L_m$ ,  $L_{k1}$ ,  $L_{k2}$ ,  $L_{k3}$ , and  $L_{kt}$ .

Fig. 9 presents the output current ripple according to the number of turns of the coupled inductors:  $N_{L1}$  ( $=N_{L2}$ ) and  $N_{L3}$ . Since the leakage inductance cannot be accurately predicted, assuming  $L_{k1}$  and  $L_{k2}$  are 10 % of  $L_m$ ,  $L_{k3}$  is 20 % of  $n_{L3}^2 L_m$  and  $L_{kt}$  is fixed at 0.3 uH for the analysis. As shown in Fig. 9, for each  $N_{L1}$  ( $=N_{L2}$ ), there is  $N_{L3}$  where the output current ripple is minimized.

Fig. 10 shows the output current ripple according to  $N_{L3}$  and  $L_{kt}$  when  $N_{L1}$  ( $=N_{L2}$ ) is 4. As demonstrated in Fig. 10, a larger value of  $L_{kt}$  tends to further reduce the output current ripple.

Fig. 11 presents the output current ripples of the proposed and conventional converters according to the load current. For reference, the output current ripple of the conventional converter can be derived from Fig. 2. When comparing the output current ripple of both converters, the following assumptions are made:  $L_{kt}$  is 0.3  $\mu\text{H}$ ,  $L_m$  is 2.512  $\mu\text{H}$ ,  $L_{k1}$  and  $L_{k2}$  are 0.2512  $\mu\text{H}$ , and  $L_{k3}$  is 0.785  $\mu\text{H}$ . The selection of the inductance values for  $L_m$ ,  $L_{k1}$ ,  $L_{k2}$ , and  $L_{k3}$  will be discussed in detail in Section IV. As shown in Fig. 11, the output current ripple of the proposed converter was much smaller than that of the conventional converter over the entire load range. Moreover, as the load current increases, the output current ripple of the conventional converter exhibits a significant increase, whereas the proposed converter shows a considerably smaller increase in the output current ripple. As a result, the proposed converter is more advantageous than the conventional converter in terms of reducing the output voltage ripple due to the reduction in output current ripple, which would directly translate into a reduction in the size of the output capacitor.

Fig. 12 shows the output current ripples for the conventional and proposed converters according to the input voltage. The inductance values of  $L_m$ ,  $L_{k1}$ ,  $L_{k2}$ ,  $L_{k3}$  and  $L_{kt}$  used to obtain the plot in Fig. 12 are as follows:  $L_{kt} = 0.3 \mu\text{H}$ ,  $L_m = 2.512 \mu\text{H}$ ,  $L_{k1}$  and  $L_{k2} = 0.2512 \mu\text{H}$ , and  $L_{k3} = 0.785 \mu\text{H}$ . These values are the same as those used in the previous analysis. Fig. 12 demonstrates that the output current ripple of the two converters is largest when the input voltage is within the middle range. Moreover, the output current ripple of the conventional converter is approximately four times larger than that of the proposed converter across all input voltage ranges.

### **(Revised part in ANALYSIS OF THE PROPOSED CONVERTER)**

#### *C. Output current ripple*

As shown in Fig. 3, the output current of the proposed converter is equal to  $i_{Lk3}$ . Furthermore, the voltage applied to  $L_{k3}$  for each mode was derived in Section II, and the effective duty cycle and commutation time were derived in A and B of Section III. Therefore, the slope of  $i_{Lk3}$  can be obtained

for each mode of operation and the output current ripple can be determined accordingly. Moreover, the slope of  $i_{Lk3}$  is influenced by  $L_m$ ,  $L_{k1}$ ,  $L_{k2}$ ,  $L_{k3}$ , and  $L_{kt}$ .

Fig. 9 presents the output current ripple according to the number of turns of the coupled inductors:  $N_{L1}$  ( $=N_{L2}$ ) and  $N_{L3}$ . Since the leakage inductance cannot be accurately predicted, assuming  $L_{k1}$  and  $L_{k2}$  are 10 % of  $L_m$ ,  $L_{k3}$  is 20 % of  $n_{L3}^2 L_m$  and  $L_{kt}$  is fixed at 0.3 uH for the analysis. As shown in Fig. 9, for each  $N_{L1}$  ( $=N_{L2}$ ), there is  $N_{L3}$  where the output current ripple is minimized.

Fig. 10 shows the output current ripple according to  $N_{L3}$  and  $L_{kt}$  when  $N_{L1}$  ( $=N_{L2}$ ) is 4. As demonstrated in Fig. 10, a larger value of  $L_{kt}$  tends to further reduce the output current ripple.

Fig. 11 presents the output current ripples of the proposed and conventional converters according to the load current. For reference, the output current ripple of the conventional converter can be derived from Fig. 2. When comparing the output current ripple of both converters, the following assumptions are made:  $L_{kt}$  is 0.3 uH,  $L_m$  is 2.512 uH,  $L_{k1}$  and  $L_{k2}$  are 0.2512 uH, and  $L_{k3}$  is 0.785 uH. The selection of the inductance values for  $L_m$ ,  $L_{k1}$ ,  $L_{k2}$ , and  $L_{k3}$  will be discussed in detail in Section IV. As shown in Fig. 11, the output current ripple of the proposed converter was much smaller than that of the conventional converter over the entire load range. Moreover, as the load current increases, the output current ripple of the conventional converter exhibits a significant increase, whereas the proposed converter shows a considerably smaller increase in the output current ripple. As a result, the proposed converter is more advantageous than the conventional converter in terms of reducing the output voltage ripple due to the reduction in output current ripple, which would directly translate into a reduction in the size of the output capacitor.

Fig. 12 shows the output current ripples for the conventional and proposed converters according to the input voltage. The inductance values of  $L_m$ ,  $L_{k1}$ ,  $L_{k2}$ ,  $L_{k3}$  and  $L_{kt}$  used to obtain the plot in Fig. 12 are as follows:  $L_{kt} = 0.3$  uH,  $L_m = 2.512$  uH,  $L_{k1}$  and  $L_{k2} = 0.2512$  uH, and  $L_{k3} = 0.785$  uH. These values are the same as those used in the previous analysis. Fig. 12 demonstrates that the output current ripple of

the two converters is largest when the input voltage is within the middle range. Moreover, the output current ripple of the conventional converter is approximately four times larger than that of the proposed converter across all input voltage ranges.

As a result, if conventional and proposed converters maintain the same output voltage ripple, the proposed converter can reduce the number of output capacitors due to a significant reduction in the output current ripple compared to the conventional Weinberg converter.

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▪ **Comment 3:**

Control scheme of the proposed converter is essential.

→ **Answer:** Thank you for your insightful comment. To regulate the output voltage of the converter described in this paper, a type 2 controller was utilized for the voltage control. Additionally, the current flowing through each switch was monitored, and a peak current control technique was applied to verify the operation of the proposed converter. The block diagram of the proposed converter control scheme is shown in Fig. R5.1 below.

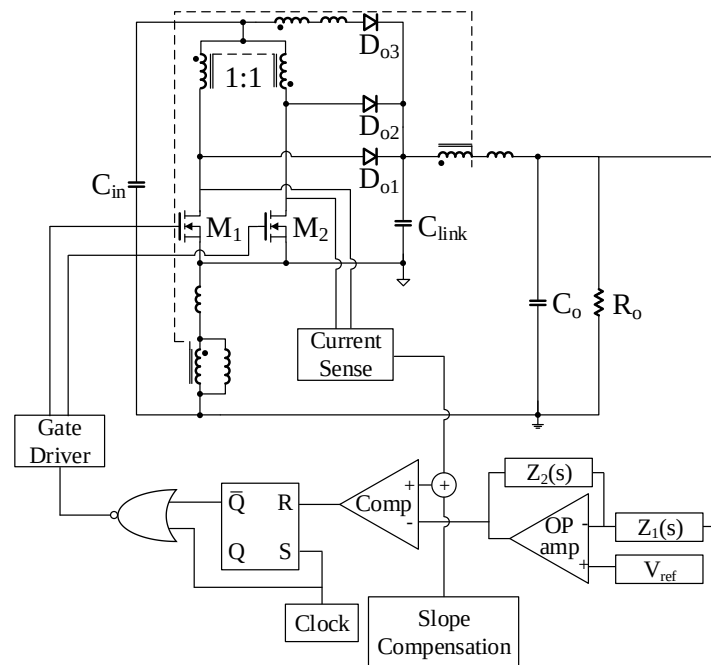


Fig. R5. 1 Block diagram of the proposed converter control scheme

According to your feedback, the control block diagram and explanation of the proposed converter have been added to Section V of this paper, as shown below.

→ [In revised manuscript]

**(Previous part in EXPERIMENTAL RESULTS)**

To verify the validity of the proposed converter, this section presents the experimental results from prototypes of both the conventional and proposed converters based on the specifications in Table IV. The prototypes of both converters are depicted in Fig. 20, the experimental key waveforms of the two converters are shown in Fig. 21, and the voltage and current waveforms of switch M1 and diodes Do1 and Do3 are shown in Fig. 22. Furthermore, Fig. 23 represents the dynamic load experimental results of both the conventional and proposed converter.

**(Revised part in EXPERIMENTAL RESULTS)**

To verify the validity of the proposed converter, this section presents the experimental results from prototypes of both the conventional and proposed converters based on the specifications in Table IV. Fig. 19 shows the control scheme of the proposed converter for experiments. As shown in Fig. 19, the proposed converter applies a type 2 controller to regulate the output voltage. Furthermore, the current flowing through switches  $M_1$  and  $M_2$  is monitored for peak current control. The prototypes of both converters are depicted in Fig. 20, the experimental key waveforms of the two converters are shown in Fig. 21, and the voltage and current waveforms of switch M1 and diodes Do1 and Do3 are shown in Fig. 22. Furthermore, Fig. 23 represents the dynamic load experimental results of both the conventional and proposed converter.

(Added Fig. 19)

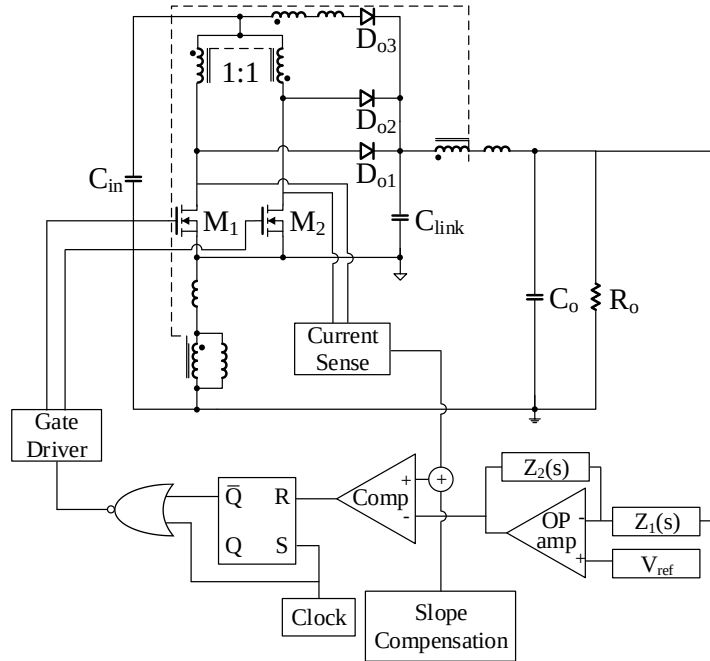


Fig. 19 Block diagram of the proposed converter control scheme

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**Reply to Reviewer 6's comments:**

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▪ **Comment:** After thorough consideration of the manuscript, along with the responses provided in the answer to the reviewers' document and the referenced circuit, it is my conclusion that the contribution and novelty of the presented circuit are deemed to be trivial. Given that the primary focus of the study lies in the topology.

J. -H. Bae, J. -E. Park and S. -K. Han, "A Battery Discharge Regulator With a Low- Output Current Ripple of an Electrical Power System in a Geostationary Satellite," in IEEE Access, vol. 11, pp. 58099-58112, 2023, doi: 10.1109/ACCESS.2023.3276246.

➔ **Answer:** Thank you for your valuable comments. The main distinction between the converter presented in the paper published in IEEE Access and the one proposed in this paper lies in the ability to reduce output current ripple without the need for an additional magnetic device. Certainly, the method of adding an inductor to the conventional Weinberg converter and the method of adding auxiliary windings to the coupled inductor of the conventional Weinberg converter may not appear significantly different from a topology perspective, as your mention. However, since each device must consider reliability for fault tolerance, the proposed converter in this paper, with one less magnetic device, may demonstrate improved reliability compared to the converter proposed in IEEE Access. Furthermore, as mentioned in this paper, the Battery Discharge Regulator (BDR) is employed in parallel in quantities of "n" according to the power capacity. Therefore, by reducing the number of magnetic devices by "n", the proposed system can improve reliability. Furthermore, the converter proposed in this paper provides design flexibility through adjustments to the turn ratio of the coupled inductor. In contrast, the converter proposed in IEEE Access designs the coupled inductor and output inductor separately. For the coupled inductor, the turn ratio is fixed at 1:1, and then, considering soft switching operation and efficiency, the turn count is determined. The output inductor, on the other hand, selects the turn count based solely on considerations of output current ripple and efficiency. In contrast, the proposed converter in this paper allows for the consideration of output current ripple, soft switching behavior of semiconductor components, efficiency, and more, based on adjustments to both turn count and turn ratio

of the coupled inductor. Therefore, the converter proposed in this paper provides a broader range for designers to create designs according to their preferences compared to the converter proposed in IEEE Access.